

Carbon Omissions:

Giulia Crippa¹, Florian Berg², and Roberto Rigobon²

¹Princeton University

²MIT Sloan School of Management

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Abstract

Corporate carbon disclosure is voluntary, and most U.S. public firms never report their emissions, leaving the majority of the cross-section outside the reach of prior empirical work. We model disclosure as a strategic decision in which equilibrium non-disclosure depends on latent emissions through a single structural parameter, γ , the *selection severity index*. Estimating this mechanism on the universe of U.S. public firms listed on the major exchanges over 2012–2023 – 87% of which never disclose – recovers the firms conventional samples omit and yields two results. First, equity markets price the act of disclosure: firms that voluntarily report earn a transparency premium of 0.225*** pp/month, orthogonal to the level of emissions. Second, strategic silence intensifies precisely where disclosure has an audience. In the cross-section, selection concentrates in high-regulatory sectors and high-institutional-ownership firms and reverses where scrutiny is weak; in the time-series, $\hat{\gamma}$ deepens through the early 2020s, coinciding with the surge in sustainable investment and a broad climate-scrutiny regime. Two quasi-experiments sharpen the mechanism: an SEC event study shows high emitters bear the brunt of regulatory shocks, and among firms that newly disclose, high emitters earn +65 bps/month more than low emitters – an effect confined to the post-2020 regime, consistent with disclosure serving as a costly credibility signal where it is most scrutinized.

Keywords: voluntary disclosure, carbon premium, selective disclosure, institutional ownership, climate regulation

JEL codes: G12, G14, Q54, M41, G18.

1 Introduction

The pricing of carbon risk in equity markets has been the subject of an active and contested empirical literature. Bolton and Kacperczyk (2021) document a robust carbon premium in vendor-estimated emissions data; Aswani et al. (2024) find no such premium in firm-reported emissions; Pastor et al. (2025) document attenuation of any premium over time. Yet all these studies, and the broader literature on which they build [TODO: add more articles], restrict attention to the self-selected sample of firms that voluntarily disclose emissions, and those estimated by data companies. The remaining 60% firms that are not included in environmental databases are absent by construction from prior asset-pricing tests, since neither observed nor vendor-estimated emissions exist for them.

This paper develops a structural model of disclosure that allows to recover the full cross-section of U.S. public firms. Specifically, we assume that firms adopt a strategic behavior when choosing whether disclosing their emissions data. According to previous literature, firms face heterogeneous incentives to disclose, shaped by regulatory exposure, reputational concerns, and potential financial consequences (Christensen et al., 2021). In particular, high-emitting firms may strategically withhold emissions data to avoid scrutiny or adverse stakeholder reactions. This assumption builds on classical results about voluntary disclosure regimes: with positive disclosure costs, silence is informative, thus rational observers interpret non-disclosure as unfavorable news (Grossman, 1981; Milgrom, 1981; Verrecchia, 1983).

In our model, the equilibrium disclosure rule is logistic in firm characteristics and latent emissions, with a single structural parameter γ – the *selection severity index* – that governs the strength of missing-not-at-random (MNAR) selection (Little & Rubin, 2019). In detail, γ decomposes into two additive channels: the private cost of revelation, and the market’s carbon-risk premium. A tightening of disclosure costs, or an increase in the attention investors pay to carbon, raises γ , deepening the adverse inference attached to silence. In other words, regulation matters because it changes the informational content of non-disclosure.

Two features of this equilibrium make it operational. First, the logistic disclosure rule implies that the emissions distribution of silent firms is an exponential tilt of the disclosed distribution (Kim & Yu, 2011), with tilt parameter γ . Second, identification requires a shifter of emissions that does not enter the disclosure decision directly (Wang et al., 2014). We build one in shift-share form, interacting the WTI crude oil price with each firm’s predetermined physical-capital intensity (Borusyak et al., 2022; Goldsmith-

Pinkham et al., 2020): oil-price movements shift the emissions of physically exposed firms more, through operating scale, without bearing on the disclosure choice itself. The interaction is essential rather than cosmetic – the oil price alone is a year-level series and is absorbed by the year fixed effects, whereas the interaction varies across firms within a year and so supplies genuine identifying variation. We recover firm-level Scope 1 emissions for the universe of U.S. public firms listed on the major exchanges – including the never-covered majority, 87% of the sample – over the time period of 2012–2023. Relative to standard missing-at-random imputations, the correction adds mass to the upper tail and raises aggregate Scope 1 emissions by about 12%, with implied carbon damages 2.5–3 times those from disclosed data under the Greenstone et al. (2023) social-cost framework. Among firms that transition to disclosure, the recovered emission distribution converge to the subsequently disclosed value in the upper tail, where vendor estimates understate by roughly 60%.

With emissions recovered for every firm, we can separate two things the disclosing-sample literature necessarily conflated: the pricing of the emissions level and the pricing of the disclosure decision. The decision dominates. The act of voluntary disclosure carries a premium of 0.225 percentage points per month, robust across sample definitions and fixed-effect structures, and orthogonal to the emissions level: adding recovered emissions to the return regression leaves the premium essentially unchanged, and the emissions coefficient is itself indistinguishable from zero. This is the resolution the disclosing-sample puzzle could not reach: the market prices transparency, not emissions. It also rationalizes the conflicting prior findings: a premium estimated on emissions levels is fragile because the level is not the priced object, while the disclosure margin that does carry information was differenced away by sampling on disclosers.

If silence is priced because an audience interprets it, then selection should bite only where an audience exists. This is the model’s sharpest testable implication, and the cross-section confirms it. The clean test holds production technology fixed and varies the audience: sorting firms into within-sector institutional-ownership terciles, selection does not merely weaken as monitoring falls away, it *reverses*. Firms in the top ownership tercile show strong strategic silence ($\hat{\gamma} = +0.19^{***}$), while those in the bottom tercile show significant reverse selection (-0.11^{***}) – higher emitters there are *more* likely to disclose – and the difference is highly significant ($p < 0.001$). Across sectors the ordering agrees: selection concentrates in Energy ($+1.39^{***}$) and Utilities ($+0.55^{***}$), the two sectors facing the most direct climate scrutiny, exceeding every other sector by roughly an order of magnitude.

The same logic operates over time. Estimated year by year, the selection severity

is statistically indistinguishable from zero throughout the 2010s and then intensifies sharply over the final three years: $\hat{\gamma}_t$ rises to +0.14*** in 2021, +0.20*** in 2022, and +0.22*** in 2023, each larger than any earlier estimate. We characterize this as a period, not a date: a cutoff-sensitivity scan rejects the null of constant γ decisively but does not locate a break, since its Wald surface rises monotonically to the edge of the trimming window and its argmax tracks the sample end rather than any event. Two independent pieces of evidence tie this intensification to a broad shift in climate scrutiny rather than to any single rule. First, the market’s response to the SEC’s March 2022 climate-disclosure proposal: among never-disclosing firms, cumulative abnormal returns load negatively on recovered emissions in tight windows around the announcement (-0.41% over $[-1, +1]$, -0.78% over $[-5, +5]$), the estimator passes a same-date placebo in a non-event year cleanly, and vendor estimates fail that placebo – evidence that the recovered emissions, not the vendor proxy, carry the pricing signal. Second, firms that transition from silence to disclosure: high emitters are repriced upward relative to low emitters by about 0.65 percentage points per month around the transition, an effect concentrated entirely in the 2020–2023 disclosure cohort and absent in a three-year-shifted placebo. Voluntary disclosure in the shadow of a prospective mandate functions as a credible commitment, and the market pays for it.

The paper contributes to three literatures. To the voluntary disclosure literature (Dye, 1985; Grossman, 1981; Milgrom, 1981; Verrecchia, 1983), we provide the first direct test of where the Milgrom– Verrecchia strategic-silence mechanism operates, using a structural MNAR framework applied to a sample (77% of public firms) that prior asset-pricing work could not study. To the carbon asset-pricing literature (Aswani et al., 2024; Bolton & Kacperczyk, 2021; Pastor et al., 2025), we extend the relevant sample from the self-selected disclosing subset to the full universe of U.S. public firms, and we identify the disclosure decision itself – rather than the level of disclosed emissions – as the channel through which carbon risk enters prices. To the missing-data and ESG-measurement literatures (Chen et al., 2025; Duan & Pelger, 2025; Kim & Yu, 2011; Olesiewicz et al., 2021; Wang et al., 2014; Yang & Kim, 2017), we provide a game-theoretic microfoundation for MNAR in carbon disclosure, quantify the resulting selection bias, and demonstrate that MNAR-corrected emissions deliver cleaner asset-pricing identification than vendor estimates.

The remainder of the paper proceeds as follows. Section ?? develops the disclosure-equilibrium model. Section ?? describes the data. Section ?? presents the imputation procedure and the resulting measurement of emissions across the full cross-section. Section ?? contains the pricing tests: the transparency premium, the boundary-conditions

evidence on where strategic silence operates, and the regulatory-anticipation regime. Section ?? reports aggregate implications. Section ?? concludes.

2 Theoretical Framework

This section develops a game-theoretic model of voluntary emissions disclosure. We begin by locating the model within the classical theory of disclosure, which supplies both the economic logic – silence is informative – and the specific frictions that allow non-disclosure to survive in equilibrium. We then specify the primitives, derive the equilibrium disclosure rule and the selection severity index γ , translate the equilibrium into an estimable correction for the missing emissions distribution, and characterize how γ responds to regulatory and market forces.

2.1 Voluntary disclosure and the informativeness of silence

The economics of voluntary disclosure¹ begins with a striking benchmark. When a privately-informed party can disclose its information at no cost, and disclosures are *verifiable* – the firm may withhold but cannot fabricate – then no information is withheld in equilibrium. The argument is the *unraveling* logic of Grossman and Hart (1980), Grossman (1981) and Milgrom (1981). A firm holding the most favorable news discloses to separate itself from the pool; its departure lowers the market’s expectation of the firms that remain silent; the best of the remaining firms then finds it optimal to disclose in turn; and the iteration continues until only the single worst type is left withholding. Rational skepticism about silence – the market assumes the worst about anything not revealed – forces full revelation. In this frictionless benchmark there is no missing data to speak of: the very possibility of concealment is undone by the inference it invites.

Systematic non-disclosure therefore has to be explained by a departure from this benchmark, and the literature offers two canonical models. The first introduces a *positive disclosure cost*. Verrecchia (1983) shows that when revealing information is costly, unraveling halts at an interior threshold: firms whose news is favorable enough to justify the cost disclose, while those below the threshold pool in silence. The withholding region is no longer a single worst type but a whole interval, and its size grows with the cost. The second departure introduces *uncertainty about whether the firm is informed*.

¹For surveys, see Verrecchia (2001) and Beyer et al. (2010).

Dye (1985) and Jung and Kwon (1988) observe that if the market cannot tell an uninformed firm (which has nothing to disclose) from an informed firm that is concealing bad news, then silence is no longer unambiguous proof of bad news, and informed firms can hide behind the uninformed. Both frictions break full unraveling and, crucially for our purposes, generate an equilibrium in which a systematic, non-random subset of firms is silent.

What unifies these models, and what we carry into the carbon setting, is that *silence is priced*. A firm that does not disclose is not treated as an average firm; it is priced at the conditional expectation of the withholding pool, which under adverse selection lies strictly below the unconditional mean. This “suspicion discount” is the economic content of non-disclosure: the market’s rational conjecture about what is being hidden, applied uniformly to everyone who hides it. In the emissions context the mapping is immediate. The private signal is the firm’s Scope 1 emissions; “bad news” is high emissions; and the cost of disclosure – regulatory scrutiny, litigation exposure, and reputational damage – plausibly scales with the quantity revealed. A firm that stays silent about its carbon footprint is pooled with, and priced like, the other silent firms, which are disproportionately high emitters.

Two modeling choices turn this qualitative logic into an estimable object. First, we make the disclosure cost increase in the *magnitude* of emissions rather than imposing a fixed cost. This replaces the binary threshold rule of Verrecchia (1983) and Dye (1985) with a smooth disclosure propensity that is logistic in latent emissions and is indexed by a single structural parameter, γ – the *selection severity index*. In the language of the missing-data taxonomy (Little & Rubin, 2019), strategic withholding is exactly a *missing-not-at-random* (MNAR) mechanism: the probability that an emissions value is unobserved depends on that very value. This is precisely the case ruled out by the missing-at-random (MAR) assumption on which conventional imputation relies, and it is the source of the bias we correct. Second, as we show below, the density of emissions among silent firms is a one-parameter exponential tilt of the disclosed density, with tilt parameter γ (Kim & Yu, 2011). The correction that undoes selection is thus fully summarized by the same γ that governs the disclosure decision – an equilibrium implication of the model, not an assumption imposed on the data.

Finally, the theory is explicit about a boundary condition that our empirical design exploits. Silence is informative only when someone *interprets* it: the suspicion discount requires an audience that conditions its beliefs on the disclosure decision. Where no such audience exists – where carbon carries little regulatory or investor scrutiny – adverse inference has no bite, and the disclosure decision is governed instead by motives

such as signaling to attract capital or idiosyncratic owner preferences, under which the sign of selection can reverse. The model therefore predicts not merely *that* MNAR selection exists but *where*: $\gamma > 0$ in environments with an interpreting audience, and $\gamma \leq 0$ where that audience is absent. This sign prediction across observable environments is the theory’s empirical fingerprint, and it is what makes the Milgrom–Verrecchia mechanism testable rather than merely invoked.

2.2 Primitives

Consider a unit continuum of firms $i \in [0, 1]$. Firm i privately assesses its Scope 1 emissions $y_i \in \mathbb{R}$, with market prior $y_i \sim \mathcal{N}(y_0, h_0^{-1})$; let ϕ denote the prior’s density.² Let $x_i \in \mathbb{R}^k$ denote firm’s i observable characteristics. The firm chooses $\delta_i \in \{0, 1\}$ (disclose/withhold) to maximize price net of disclosure costs. We work with y_i in logs throughout, so that “high emissions” refers to the upper tail of the log distribution and the Gaussian prior is the empirically relevant benchmark for the outcome model of Section ??.

Timing and information. The disclosure game unfolds over three dates. At $t = 0$, nature draws each firm’s emissions y_i from the prior and an idiosyncratic disclosure-cost shock $\tilde{\eta}_i$; the firm privately learns its own type – its emissions assessment y_i and its cost shock $\tilde{\eta}_i$ – while the market observes only the characteristics x_i and the common prior. At $t = 1$, firms simultaneously choose $\delta_i \in \{0, 1\}$. At $t = 2$, the market observes the disclosure profile, forms its posterior information set Ω , and prices each firm through the schedule below. Two features of this environment do the work in what follows. First, disclosure is *verifiable*: a firm that discloses reveals its true y_i and cannot misreport, so the only strategic margin is whether to reveal at all, exactly as in the Grossman (1981)–Milgrom (1981) tradition. Second, because firms form a continuum, no individual firm’s report moves the composition of the non-disclosure pool; the pricing of silence is therefore a fixed point of the market’s beliefs rather than a bilateral inference, which is what renders the equilibrium tractable.

We adopt the following price function³,

²We write “assesses” rather than “observes” deliberately. Producing a *verifiable* emissions figure requires measurement and, often, third-party assurance – this is the costly act the firm chooses over – whereas forming an internal assessment of whether one is a high- or low-emission firm, from directly-observed fuel and energy use, is comparatively cheap. It is this decision-relevant assessment, not an audited number, that the firm brings to the disclosure choice. Firms lacking even the internal assessment – the uninformed type of Dye (1985) – are treated in the remark following Corollary ??.

³Note that, we depart from the canonical Verrecchia (1983) variance-discount price $P(\Omega) = \mathbb{E}[y \mid$

$$P(\Omega) = V_0 - \beta \mathbb{E}[y \mid \Omega], \quad \beta > 0, \quad (1)$$

where Ω is the market's information set after observing δ_i , V_0 is a baseline value, and β is the equilibrium price impact per unit of revealed emissions, i.e. the carbon-risk premium of Bolton and Kacperczyk (2021) and Pastor et al. (2025) and it operates through revelation. When $\delta_i = 1$, then a disclosing firm is priced at its revealed emissions level y_i ; when $\delta = 0$, a silent firm is priced at the market's rational expectation over the non-disclosing pool. In particular, the value of silence equals $V_0 - \beta \mathbb{E}[y \mid \delta = 0]$.

2.3 The selection severity index

The price schedule (??) is the *benefit* side of the disclosure decision: revealing y_i moves the firm off the pooling price of silence and onto its own price – attractive for a low emitter, punishing for a high one. What disciplines the decision is the *cost* of coming forward, which we now specify. The central modeling choice, and the point of departure from the threshold models of Verrecchia (1983) and Dye (1985), is that this cost rises with the *magnitude* of emissions the firm would have to report: regulatory scrutiny, litigation exposure, and reputational damage all scale with how much carbon a disclosure reveals, so a high emitter pays more to disclose than a low emitter does. This single feature is what will make selection non-ignorable. Formally, disclosure is costly,

$$\tilde{c}_i = c_0(x_i) + c_1 y_i + \lambda \tilde{\eta}_i, \quad c_0, c_1, \lambda > 0, \quad (2)$$

where $c_0(x_i)$ is a fixed cost, c_1 is the per-unit cost of disclosing emissions, and $\tilde{\eta}_i$ a mean-zero idiosyncratic shock. The emissions dependent term $c_1 y_i$ is the engine of MNAR selection, and extends the proprietary-cost mechanism of Verrecchia (1983) and Dye, 1985, in which firms with sufficiently unfavorable private information withhold. The fixed cost $c_0(x_i)$ absorbs firm- and sector-level determinants of disclosure – measurement infrastructure, board and investor pressure, prevailing reporting norms – while c_1 isolates the marginal deterrent specific to the emissions level, the object our identification targets.

$\Omega] - \rho \text{Var}[y \mid \Omega]$ under the assumption that the penalty for silence is purely driven by adverse inference – the silent firm is priced at the pooling mean of non-disclosers in the tradition of Milgrom (1981) and Dye (1985) – rather than uncertainty aversion.

To turn this benefit– cost trade-off into a sharp disclosure rule, we impose a standard distributional structure on the idiosyncratic component.

Assumption 2.1. *The shocks $\tilde{\eta}_i$ are i.i.d. standard logistic and independent of y_i and x_i .*

Assumption ?? relies on the discrete-choice tradition of McFadden (1972); however, the qualitative results hold for any log-concave shock density with full support. [TODO: Extension in Appendix.]

Lemma 2.2 (Individual best response). *Fix the non-disclosure firms mean $\mu_{ND} = \mathbb{E}[y \mid \delta = 0]$. Under (??), (??) and Assumption ??, firm i discloses iff*

$$\tilde{\eta}_i \leq f(x_i) - \gamma y_i, \quad f(x_i) \equiv \frac{1}{\lambda}[\beta \mu_{ND} - c_0(x_i)], \quad \gamma \equiv \frac{\beta + c_1}{\lambda} > 0, \quad (3)$$

so the conditional disclosure propensity is $\mathbb{P}(\delta_i \mid x_i, y_i) = \Lambda(f(x_i) - \gamma y_i)$, where Λ is the CDF of a standard logistic.

Proof. See Appendix ??. □

Lemma ?? takes the pooling mean μ_{ND} as given. But μ_{ND} is itself an equilibrium object: the disclosure rule determines which firms stay silent, which determines the mean of the silent pool, which feeds back into the threshold $f(x_i)$ through the value of silence. Closing the model requires a μ_{ND} consistent with the disclosure behavior it induces. Theorem ?? establishes that such a fixed point exists and is unique, and – crucially – that the emissions slope γ is pinned down independently of it.

Theorem 2.3 (Existence, uniqueness, logistic propensity). *Under (??)-(??), Assumption ??, and regularity, there exists a unique equilibrium threshold f^* , and the equilibrium disclosure propensity is*

$$\mathbb{P}(\delta_i \mid x_i, y_i) = \Lambda(f^* - \gamma y_i), \quad \gamma = \frac{\beta + c_1}{\lambda} > 0. \quad (4)$$

Proof. See Appendix ??. □

Remark. γ is fixed by the slope of the disclosure inequality and is invariant to the continuum assumption and the existence/uniqueness of f^* . Multiplicity of f^* , were it to arise, would relocate the threshold but leave the identified γ unchanged.

The two components of the equilibrium propensity play distinct roles. The intercept f^* absorbs everything that shifts the disclosure decision uniformly across emission levels

– the fixed cost $c_0(x_i)$, the pooling value of silence, and any characteristic in x_i – and is a nuisance object for our purposes. The slope γ is the parameter of interest: it measures how sharply the disclosure probability falls as latent emissions rise, and it is the sole channel through which the *level* of emissions enters the decision. Its structure, $\gamma = (\beta + c_1)/\lambda$, shows that a single index aggregates two economically distinct forces – the market’s carbon-risk price β and the private per-unit cost of revelation c_1 – each normalized by the dispersion λ of the idiosyncratic shock. This is what makes γ a *sufficient statistic* for selection: two economies with different primitives but the same $(\beta + c_1)/\lambda$ generate identical disclosure patterns and identical MNAR bias. The empirical program of the paper is, in effect, the estimation of this one index and the interpretation of its variation across environments and over time.

2.4 From equilibrium to empirical correction

The equilibrium in Theorem ?? is a statement about disclosure *propensities*. To take the model to data on emissions, we need the object it implies for the *distribution* of the values that go unobserved. The next result supplies it: the density of emissions among silent firms is a one-parameter exponential tilt of the disclosed density, governed by the same γ . This is the bridge from equilibrium to estimation – it shows that the entire correction for non-ignorable selection is summarized by the selection severity index, which the propensity model of Section ?? identifies, so that no separate assumption on the missing-data mechanism is required.

Corollary 2.4 (Exponential Tilting). *Let $f_1(y | x)$ and $f_0(y | x)$ denote the conditional emission densities for disclosing and non-disclosing firms, respectively. Under (??),*

$$f_0(y | x) = f_1(y | x) \cdot \frac{\exp \gamma y}{\mathbb{E}[\exp \gamma \tilde{y} | x, \delta = 1]}. \quad (5)$$

Proof. See Appendix ??. □

Equation (??) is the exponential-tilting representation of Kim and Yu (2011), extended semiparametrically by Yang and Kim (2017); here it is derived as an equilibrium consequence rather than posited as a missing-data assumption. It applies without a mixture correction whenever every non-discloser holds the internal assessment on which the disclosure rule acts, so that silence is always a strategic choice; the remark below relaxes this and shows the relaxation reproduces exactly the missing-at-random baseline that the empirical procedure already imposes.

Remark (Uninformed firms and the MAR component). Corollary ?? treats every non-discloser as a strategic withholder. Some firms plausibly lack even the internal assessment – they have never measured their emissions and thus cannot produce a verifiable disclosure regardless of y_i , the uninformed type of Dye (1985) and Jung and Kwon (1988). Let a firm be informed with probability $\rho(x_i)$ that, conditional on x_i , is independent of y_i . Non-disclosure then pools two populations, and the silent-firm density is the mixture

$$f_0(y | x) \propto \underbrace{(1 - \rho(x)) f(y | x)}_{\text{uninformed: MAR}} + \underbrace{\rho(x) (1 - \Lambda(f^* - \gamma y)) f(y | x)}_{\text{strategic: MNAR tilt}}, \quad (6)$$

whose first term is the unconditional (missing-at-random) density of the uninformed and whose second is the exponential tilt (??) of the strategic withholders. This is the structure the empirical procedure imposes: the baseline MAR imputation of Section ?? recovers the first term and the exponential-tilt reweighting recovers the second. The pure tilt (??) is the $\rho \equiv 1$ special case; because ρ is conditioned on x but not y , the uninformed margin is ignorable and $\hat{\gamma}$ retains its interpretation as the strength of *strategic* selection.

Corollary ?? pins down the full *shape* of the correction, the entire density of unobserved emissions. For most purposes, though, what matters is a single summary of it: how far, and in which direction, selection moves the average silent firm away from the average discloser. This is the object a researcher who observes only disclosers gets wrong, and the next result characterizes it. It shows the gap has a definite sign and grows monotonically in the selection severity, so that γ orders economies not only by how they select but by how badly a naive disclosed-sample comparison understates the rest.

Proposition 2.5 (Monotone MNAR bias). *Define $\Delta\mu(\gamma) \equiv \mathbb{E}[y|\delta = 0] - \mathbb{E}[y|\delta = 1]$. Under (??), $\Delta\mu(0) = 0$, $\Delta\mu(\gamma) > 0$ for all $\gamma > 0$, and $\Delta\mu$ is strictly increasing in γ .*

Proof. See Appendix ??. □

Proposition ?? is the equilibrium analog of the upward revision in reported emissions that follows mandatory-disclosure mandates (Christensen et al., 2021): the average non-discloser is a higher emitter than the average discloser whenever strategic silence operates.

The result is sharp about *how* that gap grows. As the proof shows, the marginal effect of selection is $d\Delta\mu/d\gamma = \text{Var}_\gamma[y]$, the variance of emissions within the silent pool: selec-

tion bites harder precisely where the emissions distribution is more dispersed, because a steeper tilt then has more upper-tail mass to redistribute. When the disclosing-firm density is approximately Gaussian with variance σ_1^2 , the relationship collapses to the closed form

$$\Delta\mu(\gamma) = \gamma \sigma_1^2, \quad (7)$$

so the mean of the silent pool exceeds that of disclosers by exactly the selection severity scaled by the dispersion of disclosed emissions. The bias is not an artifact of functional form; it is a direct, monotone reading of γ , and it vanishes only in the knife-edge case $\gamma = 0$.

That knife-edge is exactly the missing-at-random benchmark. At $\gamma = 0$, silence carries no information about y : disclosing and non-disclosing firms are drawn from the same conditional distribution, and any imputation that ignores selection is unbiased. For every $\gamma > 0$, by contrast, a firm's silence is itself evidence that it sits higher in the emissions distribution, so conventional MAR imputation *understates* the emissions of non-disclosers – and understates them more, the larger is γ . This is the theoretical content behind the aggregate correction we report in Section ??: because the overwhelming majority of firms are silent, even a moderate $\hat{\gamma}$ compounds into a substantial upward revision of aggregate emissions relative to the MAR baseline. Proposition ?? guarantees that this revision is signed and monotone, so its entire magnitude is governed by the single index the propensity model estimates – which is what makes the question of the next subsection, what moves γ , the economically decisive one.

2.5 Time variation and the regulatory channel

Because the selection severity index is a function of the structural primitives, $\gamma = (\beta + c_1)/\lambda$, anything that shifts those primitives shifts the observable strength of selection. This is what allows us to read regulatory and market forces directly off the estimated $\hat{\gamma}$ – across sectors and ownership structures in the cross-section, and across time as scrutiny intensifies, among the others. The two results below make the comparative statics precise and then show how an audience-dependent benefit can flip the sign of γ , delivering the boundary condition previewed in Section ??.

Proposition 2.6 (Regulatory and market-sensitivity channels). *Under (??),*

$$\frac{\partial\gamma}{\partial c_1} = \frac{1}{\lambda} > 0, \quad \frac{\partial\gamma}{\partial\beta} = \frac{1}{\lambda} > 0, \quad \frac{\partial\gamma}{\partial\lambda} = -\frac{\beta + c_1}{\lambda^2} < 0. \quad (8)$$

Proof. Immediate from $\gamma = (\beta + c_1)/\lambda$, treating β , c_1 , λ as structural primitives. $\square$

A rise in the per-unit disclosure cost c_1 (tighter regulatory scrutiny or litigation exposure) or in the revealed-emissions price impact β (stronger carbon-risk pricing by ESG-aware investors) raises the selection severity index γ , additively and symmetrically. The empirical analog is an intensification of $\hat{\gamma}_t$ when regulatory salience or investor attention to carbon risk rises. The baseline yields $\gamma > 0$ for all primitives. Empirically observed reverse selection $\hat{\gamma} < 0$ in low-attention contexts requires a counter-force. The following extension generates it endogenously and delivers the audience boundary condition that is the empirical fingerprint of the Milgrom (1981)– Verrecchia (1983) mechanism.

Proposition 2.7 (Sign of γ and the audience boundary condition). *Augment the disclosure payoff with a transparency benefit by_i , $b \geq 0$, that scales with the emissions revealed and accrues only where a green-preferring or capital-allocating audience rewards revealed commitment. Then the equilibrium propensity retains the logistic form (??) with effective severity $\gamma_{\text{eff}} = (\beta + c_1 - b)/\lambda$, so that*

$$\gamma_{\text{eff}} > 0 \iff b < \beta + c_1, \quad \gamma_{\text{eff}} < 0 \iff b > \beta + c_1. \quad (9)$$

Proof. See Appendix ?? $\square$

Remark. Milgrom (1981) and Verrecchia (1983) predict that silence is informative only where an audience interprets the disclosure decision strategically. Proposition ?? makes this a sign condition on γ : where scrutiny dominates $\gamma > 0$ and strategic silence operates; where attention is sparse $\gamma < 0$ and selection reverses. The cross-sectional pattern of $\hat{\gamma}$ across sectors and institutional ownership is thus a direct read of the local audience structure. The proportionality of the transparency benefit to y_i is what allows b to enter the slope and flip the sign.

From theory to measurement. Taken together, the model delivers three empirical objects that organize the remainder of the paper. First, a logistic disclosure propensity whose emissions slope identifies the selection severity index γ , estimated in Section ?? . Second, an exponential-tilting correction (Corollary ??) that recovers the emissions distribution of silent firms from the disclosed distribution and the estimated γ , and hence reconstructs the full cross-section in Section ?? . Third, a sign-and-magnitude prediction for γ across audience environments (Proposition 4) that the pricing tests of Section ?? exploit as the theory’s testable fingerprint. The identification of γ from

observational data requires an excluded shifter of emissions that does not enter the disclosure decision directly; we defer that argument to the empirical framework of Section ?? and turn first to the data.

3 Data

This section describes the data sources, sample construction, and the definition of observed and missing emissions used in the analysis. We focus on Scope 1 CO₂ emissions for U.S. publicly listed firms and restrict attention to directly reported emissions in order to isolate disclosure behavior.

3.1 Sample construction

We combine firm-level emissions data from S&P Trucost Environmental with financial information from Compustat North America for U.S. publicly listed firms over 2012–2023. Macroeconomic series are drawn from FRED and include the WTI crude oil price, PPI petroleum, industrial production, and U.S. GDP. Table ?? reports the sample funnel from the raw Compustat extract to the estimation sample; we describe each step in turn, since the population we recover emissions for is itself a substantive choice rather than a housekeeping detail.

We begin with the full Compustat North America fundamentals file (161,717 firm-years) and require non-missing identifiers, fiscal-year, and core financials. We then restrict to firms listed on the three major U.S. exchanges – NYSE, NYSE American, and Nasdaq (Compustat `exchg` codes 11, 12, and 14) – which is the standard operationalization of the U.S. public-firm universe in the asset-pricing literature and the population over which our return tests are defined. This restriction is consequential and we adopt it deliberately: off-exchange listings (OTC, pink-sheet, and foreign venues) disclose Scope 1 emissions at a rate of 0.03% – seven disclosers in 21,714 firm-years – so the margin on which γ is identified is degenerate among them. Recovering their emissions would mean applying a selection correction estimated entirely on firms whose disclosure behavior they do not share, and reporting an extrapolation as a measurement.⁴ Merging to Trucost and the macro series, applying the disclosure

⁴Table ?? in Appendix ?? reports the full diagnostic. The restriction costs the paper little of its stated contribution: even within the major-exchange universe, 87% of firm-years carry no disclosed emissions, so the silent majority the paper sets out to recover lies almost entirely inside the retained sample.

classification of Section ??, and restricting to the 2012–2023 window leaves 59,996 firm-years.

Two further filters follow. To limit the influence of extreme observations we trim at the 1st and 99th percentiles of total assets within year, dropping 1,200 firm-years in the tails of the size distribution. Finally, the shift-share instrument of Section ?? requires a predetermined physical-capital exposure, so we retain firms with at least one non-missing gross PP&E-to-assets ratio in their first five sample years; because exposure is measured at the firm level and held fixed, this drops firms rather than scattered firm-years. The 7,441 firm-years lost at this step are overwhelmingly Financials (6,031) and Real Estate (1,647), sectors for which gross PP&E is sparsely populated and, in any case, for which the fuel-combustion channel underlying the instrument does not operate.

The resulting estimation sample contains 51,355 firm-years covering 6,938 unique firms, an average of 4,280 firms per year over 2012–2023. Of these firm-years, 13.0% carry a directly disclosed Scope 1 figure; the remaining 87% are the firms the paper recovers.

We focus exclusively on Scope 1 emissions, defined as direct emissions from sources owned or controlled by the firm under the Greenhouse Gas Protocol. Scope 2 and Scope 3 emissions are excluded due to substantial measurement challenges, heterogeneity in reporting standards, and the risk of double counting (Cheema-Fox et al., 2021). Restricting attention to Scope 1 emissions allows for a more precise assessment of corporate disclosure behavior and limits contamination from upstream or downstream reporting practices.

Table ?? provides annual summary statistics. These describe the sample after the asset trim but before the predetermined-exposure requirement (58,796 firm-years, averaging 4,900 firms per year), since that final filter is an instrument-construction step rather than a feature of the data; the estimation sample of 51,355 firm-years is the one used throughout Section ?. The number of covered firms declines gradually over the sample period, consistent with broader trends in U.S. equity markets, while total market capitalization increases substantially, particularly after 2020. Aggregate Scope 1 emissions remain relatively stable over time, whereas average emissions intensity declines markedly, reflecting a combination of changes in firm composition, scale effects, and improvements in emissions efficiency.

3.2 Disclosure groups

A central design choice in this paper is how to classify emissions observations as observed versus missing. In addition to firm-reported emissions – sourced from sustainability reports, CDP disclosures, or direct company communication – Trucost provides model-based estimates constructed using industry benchmarks such as the and engineering assumptions. Because these estimates do not reflect firms’ disclosure choices, we treat them as missing for the purposes of this study.

In addition, we classify as missing the emissions of firms that appear in Compustat but are not covered by Trucost Environmental (by left-merging the Trucost data onto the full Compustat universe). This choice expands the analysis to a broader cross-section of U.S. public firms and ensures meaningful representation of the overall economy. In contrast, related studies typically restrict attention to firms with emissions coverage or rely on vendor-provided estimates, thereby excluding a substantial fraction of non-reporting firms (Chen et al., 2025; Duan & Pelger, 2025; Olesiewicz et al., 2021). Table ?? reports summary statistics separately for firms with observed emissions and firms with missing emissions.

Table 1: **Scope 1 emissions data groups.** Classification of Scope 1 emissions observations by Trucost data source and MNAR treatment. *Disclosed* firms report emissions directly; *estimated* values are Trucost vendor constructions that do not reflect a firm reporting choice; *none* are Compustat firms with no Trucost coverage. The latter two are treated as missing for the disclosure-decision analysis.

Group	Definition	Treatment
Disclosed	Firm-reported: • Sustainability/CSR reports • CDP • Annual reports / 10K-s	Observed
Estimated	Trucost vendor estimate	Missing
None	No Trucost coverage	Missing

3.3 Patterns in disclosure

To better model disclosure, it is of paramount importance to analyze fundamental patterns about what firms disclose, when disclosure occurs, and how coverage evolves across firm characteristics. Understanding these mechanisms provides essential foundations for subsequent modeling assumptions, and clarifies the extent to which observed emissions reflect reporting behavior rather than underlying production technology.

Figure ?? summarizes patterns in Scope 1 emissions disclosure over time, across sectors, and by firm size. The top panel documents a steady increase in the fraction of firms that disclose emissions directly, rising from approximately 5% in 2012 to a peak near 21% in 2022 (about 19% in 2023). Over the same period, the share of firms for which Trucost provides estimated emissions expands more sharply, particularly after 2016, reflecting both expanded data coverage and heightened attention to climate reporting following the Paris Agreement. Despite these trends, a substantial fraction of firms continue to have missing emissions data throughout the sample period.

The middle and bottom panels focus on cross-sectional patterns in 2023. The middle panel reports disclosure rates by sector and reveals pronounced heterogeneity across industries. Utilities, Materials, and Energy exhibit the highest direct-disclosure rates, while Communication, Health Care, and Real Estate display substantially lower coverage. These differences are consistent with sectoral variation in emissions intensity, regulatory exposure, and stakeholder scrutiny: the most energy-intensive sectors are among the leading disclosers of Scope 1 emissions.

The bottom panel reports disclosure rates by firm-size decile, measured by total assets. Larger firms are substantially more likely to disclose emissions or to be covered by Trucost estimates, whereas coverage among smaller firms remains limited. In the bottom two deciles, fewer than 25% of firms have any emissions data available, and direct disclosure is essentially absent. In contrast, firms in the top deciles exhibit total coverage exceeding 50%, with direct reporting accounting for a large and rising share.

Together, these patterns highlight systematic variation in emissions disclosure across time, sectors, and firm size, underscoring the importance of accounting for selective disclosure in empirical analyses. In the next section, we describe the empirical framework used to address this selection when imputing missing emissions.

4 Recovering Latent Emissions

We recover Scope 1 emissions for every firm in the Compustat universe by correcting the disclosed-firm distribution for non-ignorable selection. Guided by the model of Section ??, the procedure has five steps: (i) an outcome model fit on disclosers; (ii) a baseline missing-at-random (MAR) imputation of non-disclosers from that model; (iii) a logistic disclosure model whose emissions slope is the selection severity index γ ; (iv) the exponential-tilt correction of Corollary ??, applied to the baseline draws; and (v) Monte Carlo propagation of imputation uncertainty. This section sketches each step

figs/fig1_disclosure.pdf

Figure 1: **Scope 1 emissions disclosure composition.** Each bar partitions firms into three mutually exclusive groups – *disclosed* (firm-reported emissions; dark blue), *estimated* (Trucost vendor estimate; light blue), and *missing* (Compustat firms with no Trucost coverage; dashed outline) – as a share of firms summing to 100% within each bar. **Panel A** plots the composition by year over 2012–2023; **Panel B** the 2023 composition by GICS sector; and **Panel C** the 2023 composition by firm-size decile (asset quantiles, smallest to largest). Data are from S&P Trucost Environmental merged onto the Compustat universe of U.S. public firms.

and reports what it delivers; the granular choices – functional forms, the imputation draw, the formal exclusion restriction, and the Monte Carlo algorithm – are collected in Appendix ??.

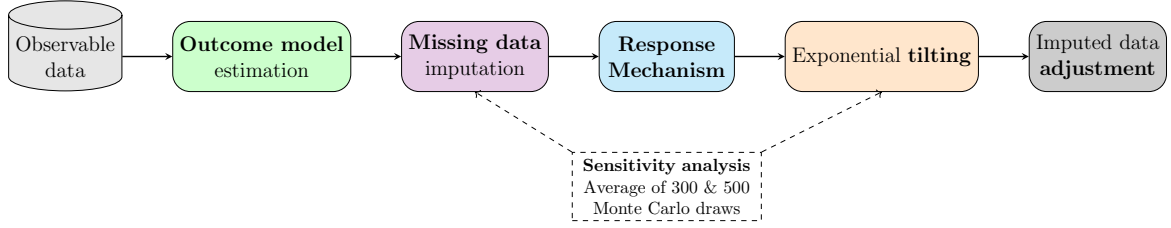


Figure 2: Workflow of the missing data imputation methodology. The procedure consists of five main steps: (1) outcome modeling based on observable data; (2) imputation of missing values; (3) estimation of the response mechanism; (4) application of exponential tilting; and (5) adjustment of the imputed values. Additionally, a sensitivity analysis is conducted using Monte Carlo simulations with 300 and 500 random draws, respectively.

4.1 Outcome model

The first step estimates the conditional distribution of emissions among disclosers with a lognormal model,

$$y_{it} = \eta_0 + \eta_1 \log \text{assets}_{it} + \eta_2 z_{it} + \mu_{j(i)} + \tau_t + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2), \quad (10)$$

where y_{it} is log Scope 1 emissions of firm i in year t , assets_{it} is total assets (a proxy for production scale, without the valuation-driven volatility of market capitalization), z_{it} is the oil-price exposure shifter used for identification (Section ??), and $\mu_{j(i)}$ and τ_t are GICS industry and year fixed effects, respectively. We use total assets to proxy production scale without valuation-driven volatility. Fit on disclosers, the model recovers the production-side signal in emissions – scale, industry technology, aggregate conditions – and supplies the conditional density $f_1(y \mid \mathbf{x})$ from which we draw baseline MAR imputations for non-disclosers (the draw is given in Appendix ??). These draws treat non-disclosers as if $f_0(y \mid \mathbf{x}) = f_1(y \mid \mathbf{x})$ – the MAR benchmark that Proposition ?? shows fails whenever $\gamma \neq 0$; the remaining steps correct it.

Table ?? presents the estimated coefficients from our model. All covariates, with the exception of a few year indicators, are statistically significant at the 5% level. The model explains a substantial portion of the variance in reported CO₂ emissions, with an adjusted R^2 of 0.81.

Firm size, proxied by log assets, shows a strong positive association with Scope 1 emissions. The coefficient estimate of 0.97 suggests that a 1% increase in firm size is associated with approximately a 0.97% increase in emissions, scaling almost linearly. The exposure shifter enters positively and strongly ($\hat{\eta}_2 = 0.19$, $t \approx 32$): among physically exposed firms, emissions co-move with the oil price, as expected when fuel-combustion

operations scale with energy costs. The coefficient is a relevance statistic rather than a structural parameter — it is the first-stage requirement that the instrument shift emissions — and its strength ($t > 30$) rules out weak-instrument concerns.

Industry differences in emissions are substantial. Differentials are estimated relative to the omitted industry, and the great majority are significant at the 1% level. The largest positive differentials fall on the most physically fuel-intensive industries: Passenger Airlines and Marine Transportation (about +2.3), Electric and Multi-Utilities (+1.5 to +1.8), Chemicals and other Materials (+1.0 to +1.7), and Oil, Gas & Consumable Fuels. The most negative are the asset-light service industries: Capital Markets, Consumer Finance, and Insurance cluster near -5 , and Software, IT Services, Interactive Media, and Professional Services lie between -3.5 and -4.2 . The finer resolution also reveals contrasts invisible at the sector level: within Utilities, Electric Utilities load strongly positive (+1.8) while Water Utilities are sharply negative (-3.0). Overall, this pattern is consistent with the broad distinction between more emission-intensive (*brown*) and less emission-intensive (*green*) sectors, reflecting structural differences in business operations and energy use.

The year fixed effects from the indicate a clear and statistically significant downward trend in reported Scope 1 emissions conditional on firm characteristics over the past decade. Using 2012 as the reference year, the coefficients for subsequent years become increasingly negative, with effects growing in magnitude and statistical significance over time. While early-year coefficients (2013–2016) are small and not statistically different from zero, the decline becomes significant starting in 2018. By 2023, the coefficient reaches -0.69 , implying that, all else equal, reported emissions in 2023 were approximately 50% lower than the 2012 baseline. This sustained decline likely reflects a combination of structural changes, including tightening climate regulation, increased stakeholder scrutiny, and improvements in emissions measurement and reporting. The accelerating pace of change after 2018 may also coincide with major global climate initiatives and improved emissions tracking mechanisms. These time effects offer strong evidence that the landscape of corporate emissions disclosure has evolved materially over the time frame considered.

Following our methodology, the estimated outcome model is used to generate plausible values for missing emissions data by randomly drawing from the predictive distribution in Equation (??). This forms the first step of the imputation process, to obtain a complete dataset that will be utilized to model the response mechanism.

4.2 Disclosure mechanism and the selection severity index

After imputing scope 1 emissions, we proceed to estimate the response propensity for each company, leveraging the complete covariates $\mathbf{x}_1$ and the dependent variable y . We model the disclosure decision with a logistic response mechanism in firm characteristics and emissions,

$$\mathbb{P}(\delta_i = 1 \mid \mathbf{x}_{1i}, y_i) = \frac{\exp\{f(\mathbf{x}_{1i}) + g(y_i)\}}{1 + \exp\{f(\mathbf{x}_{1i}) + g(y_i)\}}, \quad (11)$$

the empirical counterpart of the equilibrium propensity of Theorem ???. Our primary specification is fully parametric, with f and g linear, so $g(y_i) = -\gamma y_i$ and the coefficient on y_i estimates $-\hat{\gamma}$ directly. This is the specification the model implies: the disclosure cost is linear in emissions, so the equilibrium propensity is linear in y (Lemma ??). A semiparametric variant with penalized splines relaxes that linearity as a robustness check (Yang & Kim, 2017); its specification and results are in Appendix ??.

Identification of the emissions slope is the crux: y_i is unobserved for exactly the non-disclosers the model describes, so it cannot come from the disclosing sample alone. Identification rests on an *excluded shifter* of emissions (Wang et al., 2014), and we build one in shift-share (Bartik) form – an aggregate shock interacted with predetermined firm-level exposure (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020):

$$z_{it} = \log(\text{WTI}_t) \times \text{exposure}_i, \quad (12)$$

where WTI_t is the crude oil price and exposure_i is firm i 's predetermined physical-capital intensity: gross property, plant, and equipment over total assets, averaged over the firm's first (up to) five non-missing sample years and then held fixed. Averaging the first few years rather than taking a single year reduces the influence of one anomalous observation on a firm-level weight that is applied throughout the panel; because the exposure is time-invariant by construction, all within-firm time variation in z_{it} comes from the oil price. Because z_{it} varies across firms *within* a year, it is not collinear with the year fixed effects, and so – unlike the aggregate oil price alone, which the year effects absorb – it supplies identifying variation present in the emissions equation but excluded from the disclosure equation. It is *relevant*: oil-price movements shift the emissions of physically exposed firms more, and z_{it} enters the outcome model (??) strongly (coefficient 0.19, $t \approx 32$). *Exclusion* rests on the exogeneity of the aggregate shock (Borusyak et al., 2022): the crude oil price is exogenous to any individual firm's disclosure decision, and firms differ only in *predetermined* exposure, so z_{it} shifts a firm's

emissions level but has no direct channel to its choice to disclose, conditional on size and sector. Appendix ?? states the exclusion restriction formally, and Appendix ?? reports the estimates obtained when the WTI shock is replaced by U.S. real GDP.

The estimate is signed as the model predicts (Table ??): $\hat{\gamma} = 0.199$ ($B = 500$; 0.195 at $B = 300$), positive and significant at the 1% level – higher emitters are systematically less likely to disclose. This is direct empirical support for non-ignorable disclosure and a measured value of the structural selection severity of Theorem ?. Coefficients are standardized, so magnitudes read as the effect of a one-standard-deviation move evaluated at the sample disclosure rate of 13.0%. A one-standard-deviation increase in firm size raises the disclosure probability from 13.0% to approximately 49%: larger firms are far more likely to report, which is the dominant force in the disclosure decision and the reason $f(\mathbf{x})$ must absorb size before the emissions slope is interpretable. Conversely, a one-standard-deviation increase in Scope 1 emissions lowers the disclosure probability from 13.0% to approximately 10.9%, a relative decline of about 16%, consistent with higher-emitting firms withholding their environmental performance.

The emissions coefficient carries larger sampling uncertainty than the other covariates (standard error 0.074, versus 0.028 on size), which is expected: y_i is imputed for the non-disclosers whose behavior identifies γ , and the Monte Carlo propagates that imputation uncertainty into the slope. Carrying the 95% interval through the logistic transformation, the implied disclosure probability at +1 SD of emissions ranges from about 9.6% to 12.4% – everywhere below the 13.0% base rate, so the *sign* of the selection effect is stable across the interval even though its magnitude is not sharply pinned down. This is the appropriate degree of confidence to attach to the pooled estimate: the direction of strategic silence is firm, its exact strength less so, and Section ?? shows that the pooled number in any case averages over sectors and ownership structures whose selection severities differ by an order of magnitude and in sign.

4.3 The tilting correction and its distributional effect

With $\hat{\gamma}$ estimated, we correct the baseline imputations by the exponential tilt of Corollary ??, reweighting each draw by

$$w_i \propto \exp\{\hat{\gamma} \hat{y}_i\}, \quad (13)$$

so that firms with lower disclosure probabilities – the higher emitters – receive more weight, restoring the upper-tail mass that selection removes. The idea is that fore-

casted emissions from the outcome model are scaled based on the inverse likelihood of disclosure: the less likely a firm is to report, the greater the upward adjustment. However, the final tilt does not depend only on disclosure probability but also on each company’s size, business sector, and imputed emissions. Hence, a low reporting probability does not always result in a large adjustment.

The correction is economically large (Table ??). Relative to the MAR baseline mean of 957,284 tCO₂, the fully parametric tilt raises average imputed Scope 1 emissions to 1,075,583 tCO₂, an uplift of 12.4% ($B = 500$; 11.9% at $B = 300$). The semiparametric response model, which relaxes the linearity of $g(y)$, produces a larger correction of 19.1% (1,139,831 tCO₂); we report the fully parametric figure as the headline because it is the specification the model implies, and read the semiparametric number as an upper bound that indicates the correction is not an artifact of the linear functional form. Under both, the added mass is concentrated in the upper tail: the standard deviation of imputed emissions roughly doubles (46,054 to 86,510 under FP), which is the signature of a tilt that redistributes weight toward high emitters rather than shifting the whole distribution. Because the overwhelming majority of firms are silent, this compounds into a substantial upward revision of aggregate emissions (Section ??): conventional MAR imputation understates corporate emissions precisely where the level matters most for aggregate damages.

4.4 Validation: firms transitioning to disclosure

A fundamental challenge in non-ignorable missingness settings is the absence of ground truth for validating imputed values. However, a subset of firms in our sample transition from non-disclosure to disclosure over time, providing an opportunity for external validation. Specifically, we identify 633 companies for which emissions were initially estimated, and then disclosed in subsequent years. For these firms, we compare emissions imputed in the year *prior* to disclosure with the subsequently reported values. This is a calibration diagnostic, not identification: the transition firms play no role in estimation.

The MNAR-corrected imputations track the disclosed ranking more closely than the vendor estimates (Spearman $\rho = 0.86$ versus 0.81 for Trucost), and the gain concentrates exactly where it matters. Among the highest-emitting quartile, the MNAR prediction is essentially unbiased (mean log error +0.07), whereas Trucost understates emissions by about 0.74 log units – roughly 50% – precisely the upper-tail firms that dominate aggregate emissions. Both methods overstate the smallest emitters, an un-

avoidable feature of conditional-mean imputation for near-zero values, but those firms are negligible for aggregates; the accuracy that matters for aggregate emissions and damages is in the tail, where the correction excels (Figure ??). This pattern is consistent with our framework’s design, which accounts for strategic non-disclosure incentives that are more prevalent among large emitters. Given that these firms contribute disproportionately to aggregate pollution, the reduction in high-end underestimation is critical for accurately estimating the total average emissions and associated carbon damages.

Together, these steps delivers a Scope 1 emissions figure for every firm in the Compustat universe, including the majority that never disclose. Relative to Trucost estimates the recovered distribution is materially heavier in the upper tail, and the transition-firm validation confirms that the correction bites where it should – among the high emitters that dominate aggregates. With emissions in hand for disclosers and non-disclosers alike, we can finally separate two objects the prior literature was forced to conflate: the pricing of the emissions *level* and the pricing of the disclosure *decision*. Section ?? turns to that separation – and to the boundary conditions under which strategic silence is priced at all.

5 Pricing the Disclosure Decision

This section contains the central empirical results of the paper. We organize the evidence around the lead story: the market prices the disclosure decision itself (Section ??); strategic silence activates exactly where voluntary disclosure theory predicts an audience exists (Section ??); and the mechanism intensifies in a regulatory-anticipation regime that emerges around 2020 (Section ??).

5.1 Specification

Throughout, we regress monthly firm returns (in percentage points) on the recovered emissions and a disclosure indicator, with controls following Bolton and Kacperczyk (2021): log market capitalization, log book-to-market, momentum, return volatility, leverage, investment/assets, ROE, log PPE, sales growth, EPS growth, and the Herfindahl index. Standard errors are clustered by firm. We report results under four fixed-effect structures – time only; time and industry; time and firm; and a stable-sectors subsample – so that the transparency premium can be read against

progressively more demanding controls.

5.2 Diagnostic: BK replication

Before turning to the lead results, we replicate Bolton and Kacperczyk (2021) on the disclosing and disclosed-plus-estimated samples using observed, MNAR-corrected, and MAR-imputed emissions. The replication has three purposes: (i) confirm the Aswani et al. (2024) null in the disclosed universe; (ii) demonstrate that replacing observed with MNAR-corrected emissions leaves disclosed-sample coefficients unchanged; (iii) show that MNAR-corrected and MAR-imputed coefficients are similar within the BK working sample, so that the tilting itself is not mechanically driving downstream pricing results.

[TODO: Insert BK replication table (5 columns: BK Sample Observed, BK Sample MNAR, BK Sample MAR, Full Universe MNAR, Full Universe MAR).]

Emissions intensity. The raw-emissions null is specific to levels. In the intensity specification advocated by Aswani et al. (2024) – log Scope 1 scaled by revenue, which removes the mechanical size correlation that drives the levels regression – a significant negative premium emerges, and the MNAR correction *strengthens* rather than attenuates it: the coefficient moves from -0.117^{***} on observed emissions to -0.170^{***} under MNAR correction, and is -0.169^{***} on the full Compustat universe (robust to scaling by assets, -0.078^{**} on the full universe). Higher carbon-*intensity* firms earn lower returns; selection correction sharpens the relationship rather than diluting it, the opposite of what mechanical tilting would produce. The intensity premium is a statement about emissions per unit of output, distinct from the transparency premium of Section ??, which prices the disclosure decision itself and is orthogonal to the level of emissions.

[TODO: Insert intensity table: `returns_bk_intensity.tex`.]

Vendor estimates and spurious pricing. The contrast between firm-reported and vendor-estimated emissions extends from the event study to the cross-sectional pricing regression. On the vendor-estimated subsample, Trucost emissions carry a significant positive premium ($+0.059^{**}$) that disappears entirely once emissions are replaced with the MNAR-corrected values (0.008, insignificant). This reconciles the Bolton and Kacperczyk (2021)–Aswani et al. (2024) divergence directly: the premium documented

in vendor-inclusive samples is a property of the vendor estimation procedure, not of firm emissions, and it does not survive a selection-corrected measure. The result parallels the Trucost placebo failure in the SEC event study (Section ??), where vendor estimates produce significant wrong-signed coefficients on a date with no climate-policy news.

[TODO: Insert vendor-sample table: `returns_estimated.tex`.]

Framing. This subsection is a diagnostic, not a pricing claim.

5.3 The transparency premium

- Full sample, monthly returns on a Disclosed dummy and MNAR emissions, jointly and separately.
- **Result:** $+0.225^{***}$ **pp/month premium for the act of voluntary disclosure.** The coefficient is unchanged at 0.225^{***} when MNAR emissions are added and is identical under MAR-imputed emissions; the Scope 1 coefficient is effectively zero (≈ 0.006 , insig).
- Robust across all sample specifications. Standard errors ≈ 0.053 .

Interpretation. The transparency premium is the empirical analog of Milgrom (1981) and Verrecchia (1983): voluntary disclosure is rewarded because silence is penalized. The transparency premium and the MNAR-emissions coefficient are orthogonal in their return implications – the market prices the disclosure *decision*, not the level of what is disclosed.

[TODO: Insert main transparency-premium table: `returns_main.tex`.]

5.4 Boundary conditions: where strategic silence operates

Voluntary disclosure theory predicts that strategic silence is informative only when an audience interprets it as bad news. Proposition ?? makes this a sign condition: $\gamma_{\text{eff}} > 0$ where scrutiny dominates, $\gamma_{\text{eff}} < 0$ where a transparency benefit dominates instead. We test it by re-estimating the emissions slope separately within observable environments that proxy for the presence of an audience, holding sector and year fixed

effects throughout.⁵

Two restrictions discipline the exercise, and we state them before reporting results. First, γ is only identified where disclosure actually varies: in sectors where almost no firm reports, the logit is quasi-separated and the estimate diverges toward an artifact of the separation rather than a measure of disclosure behavior. We therefore report $\hat{\gamma}$ only for sectors whose disclosure rate exceeds 10%, which excludes Financials, Real Estate, Communication Services, and Health Care.⁶ Second, the threshold is specified in advance of inspecting the estimates, and it is the same rule the sample construction of Section ?? applies to off-exchange firms.

Ownership channel. The sharpest test is within-sector institutional ownership, which varies the audience while holding the technology of production fixed. Sorting firms into within-sector IO terciles and estimating one emissions slope per tercile, the pattern is exactly the sign flip the model predicts:

- High IO: $\hat{\gamma} = +0.19^{***}$ – strategic silence.
- Mid IO: $\hat{\gamma} = -0.06^{***}$.
- Low IO: $\hat{\gamma} = -0.11^{***}$ – reverse selection.
- Difference, high – low: $+0.30$, $p < 0.001$.

The economic content is not that selection is *weaker* without monitoring but that it *reverses*. Among firms in the bottom ownership tercile, higher emitters are significantly *more* likely to disclose; only under concentrated institutional ownership does the emissions slope turn positive and silence become strategic. This is Proposition ?? operating at the boundary $b = \beta + c_1$: where no audience penalizes revealed emissions, the transparency benefit b dominates and the sign of γ_{eff} flips. A theory predicting only attenuation would be consistent with a weaker high-IO effect; it would not predict that the low-IO slope is significantly negative.

⁵Throughout this subsection $\hat{\gamma}$ is reported in the sign convention of the model, so that $\hat{\gamma} > 0$ denotes strategic silence (higher emitters disclose less) and $\hat{\gamma} < 0$ denotes reverse selection. Emissions are standardized within sector, so magnitudes are log-odds per within-sector standard deviation and are comparable across groups.

⁶The excluded sectors' unrestricted estimates are large and negative (-38 for Financials, -11 for Real Estate) with correspondingly wide intervals – the numerical signature of quasi-separation, not evidence of extreme selection. Appendix ?? reports the disclosure rate of every sector alongside its unrestricted estimate so the reader can see what the threshold removes and why.

Sectoral channel. Re-estimating by GICS-11 sector orders the retained sectors as the mechanism implies, with selection concentrated in the carbon-salient ones:

- *Strong selection:* Energy (+1.39^{***}) and Utilities (+0.55^{***}) – the two sectors facing the most direct climate-regulatory and ESG scrutiny, and by a wide margin the largest estimates in the cross-section.
- *Weak selection:* Information Technology (+0.15^{***}), Consumer Staples (+0.10^{*}), Materials (+0.10^{*}), Consumer Discretionary (≈ 0 , insignificant).
- *Reverse selection:* Industrials (−0.22^{***}).

The ordering is informative – Energy and Utilities exceed every other sector by roughly an order of magnitude – but we read the sectoral evidence as weaker support for the boundary condition than the ownership evidence, for two reasons. First, sector is a coarse proxy for audience: it bundles regulatory exposure with production technology, measurement infrastructure, and reporting norms, so a sectoral $\hat{\gamma}$ need not isolate the audience channel. Second, the one sector with significant reverse selection, Industrials, is not plausibly a low-scrutiny environment, which is what a naive reading of Proposition ?? would require.

We flag rather than explain away the Industrials result. One candidate mechanism is measurement-driven and testable: the EPA’s Greenhouse Gas Reporting Program mandates reporting above 25,000 tCO₂e, and Industrials straddles that threshold, so within the sector the high emitters are disproportionately the firms for which an emissions figure already exists and is public. That would generate a positive association between emissions and disclosure without any change in the strategic motive. Consistent with this reading, Energy and Utilities – whose firms sit almost entirely above the threshold, leaving no within-sector variation in mandated coverage – show the strongest *strategic* selection instead. We regard this as a conjecture: it predicts that the Industrials slope is concentrated among firms near the reporting threshold, which we have not yet tested. [TODO: Split Industrials at the GHGRP 25,000 tCO₂e threshold and report whether the positive slope concentrates among threshold-crossers.]

Interpretation. Where firms face external attention – through concentrated institutional ownership or direct climate-regulatory exposure – high emitters strategically hide, and selection bites. Where attention is sparse, the disclosure decision is dominated by alternative motives (signaling to attract capital, idiosyncratic owner preferences, measurement capacity), and the selection pattern reverses. The cross-sectional

heterogeneity is not a complication for the theory; it is the theory’s empirical fingerprint, and the ownership channel delivers it in the sharp form the model predicts – a change of sign, not merely of magnitude.

[TODO: Insert sector $\hat{\gamma}$ and IO $\hat{\gamma}$ figures ([gamma_by_sector_fullwindow_tilt_test.pdf](#), [gamma_by_io_fullwindow_tilt_test.pdf](#)), with signs flipped to the model’s convention.]

5.5 The regulatory-anticipation regime

Three independent pieces of evidence place the intensification of carbon-disclosure relevance in the early 2020s: a year-by-year selection-severity trajectory, an SEC event study, and a transition-firm difference-in-differences. Each addresses a different margin and frequency, and we read them as parallel rather than causally linked.

5.5.1 The trajectory of $\hat{\gamma}_t$

We estimate the selection severity year by year from the propensity model alone – distinct from any return regression – by interacting emissions with year indicators, and ask when it intensifies. Figure ?? reports $\hat{\gamma}_t$ scaled by the within-year standard deviation of emissions, so each point is the log-odds change from a one-standard-deviation move in emissions in that year.

[TODO: Regenerate this figure with $\hat{\gamma}$ in the model’s sign convention (multiply by -1 ; positive = strategic silence) so the plotted signs match the text and Proposition ?. Same for the sector and IO figures.]

The trajectory is flat and then steep. From 2013 through 2020, $\hat{\gamma}_t$ is small and, with the single exception of 2018 ($+0.085^*$), statistically indistinguishable from zero: over most of the 2010s we cannot reject that disclosure is unrelated to the emissions level once size, sector, and year are controlled for. The final three years break sharply from this: $\hat{\gamma}_{2021} = +0.142^{***}$, $\hat{\gamma}_{2022} = +0.202^{***}$, and $\hat{\gamma}_{2023} = +0.222^{***}$, each significant at the 1% level and each larger than any earlier estimate.⁷ By the end of the sample, a one-standard-deviation increase in emissions moves the log-odds of disclosure by more than two and a half times what it did in the largest pre-2021 year.

⁷The first sample year is an exception to the flat early pattern ($\hat{\gamma}_{2012} = +0.067^{**}$). We do not lean on it: 2012 is the first year of Trucost coverage, when the disclosing sample is both smallest and least representative, and the estimate is not part of any monotone pattern with adjacent years.

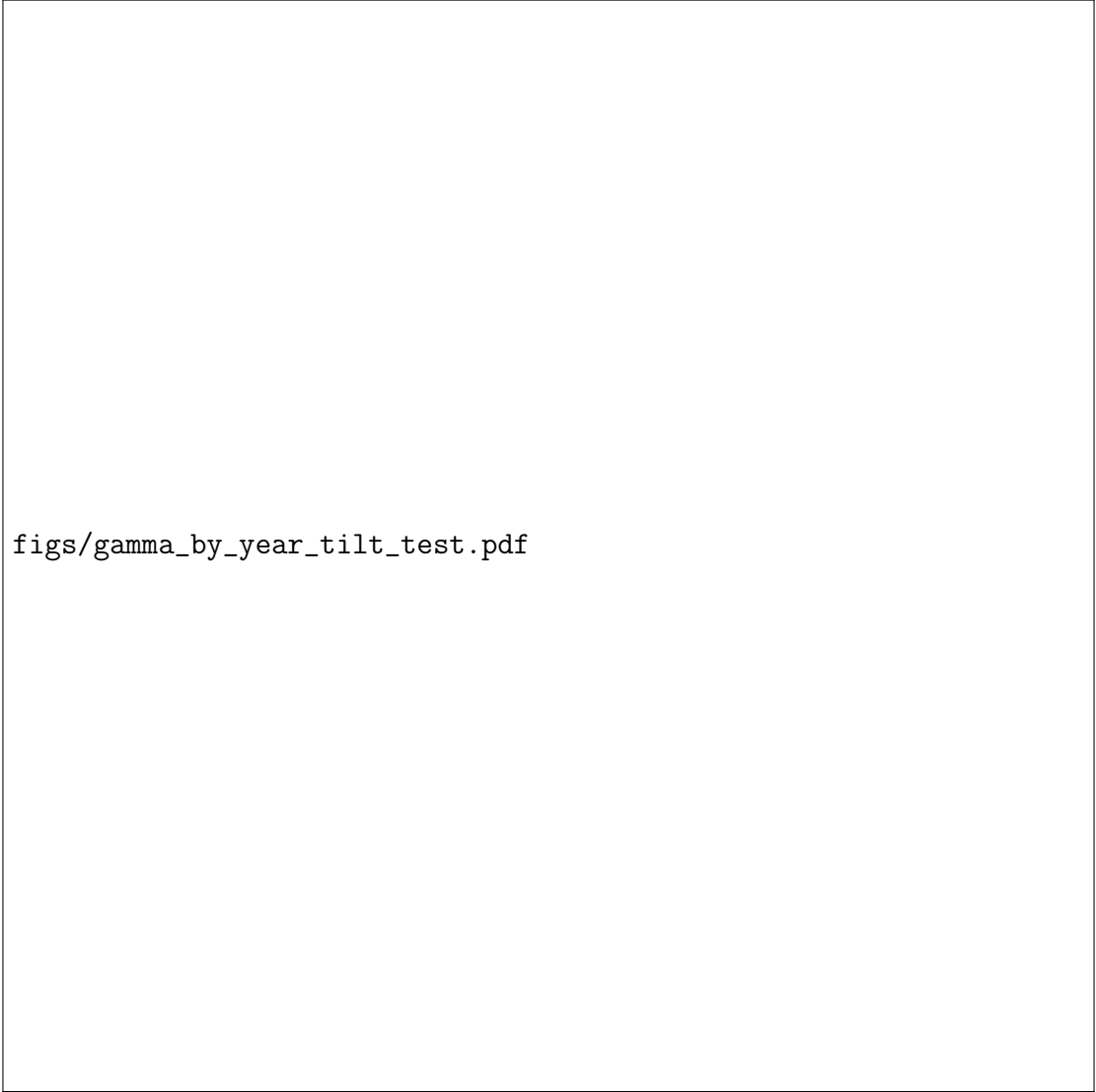


Figure 3: **Selection severity by year.** Estimated $\hat{\gamma}_t$ from a propensity model with emissions $\times$ year interactions, scaled by the within-year standard deviation of emissions, with 95% confidence intervals. Positive $\hat{\gamma}_t$ indicates strategic silence: higher emitters are less likely to disclose. The specification includes size, sector, and year fixed effects; 2012 is the first year of Trucost coverage. Stars denote significance at the 10%, 5%, and 1% levels.

The intensification is not a structural break, and we do not date one. It is natural to ask whether these three years reflect a discrete regime change at an identifiable date, and we tested for one with a cutoff-sensitivity (sup-Wald) scan over candidate split years, following **andrews1993tests<empty citation>** with 15% trimming. The test rejects the null of constant γ decisively – the maximized Wald statistic of 21.3 exceeds the 1% critical value – but it does *not* locate a break, and the reason is visible in the shape of the scan rather than in its argmax. The Wald statistic rises monotonically across every candidate cutoff (0.4 at 2014, 5.0 at 2016, 10.2 at 2018, 16.1 at 2020, 21.3 at 2022) and is still rising when the trimming window ends. A genuine break produces a Wald surface that peaks at the break date and declines on either side; a surface that is monotone to the boundary is the signature of a continuous change, or of a break at or beyond the edge of the admissible range.

The argmax is accordingly an artifact of where the scan stops rather than a feature of the data. Dropping 2023 and re-running moves the last admissible cutoff from 2022 to 2021, and the argmax follows it exactly, to 2021, while the surface remains monotone. Any date we reported from this procedure would track our sample end rather than any economic event, so we report none. We note also that the conventional p -values from such a scan are not interpretable at the argmax, since searching over cutoffs invalidates the χ^2 null – which is precisely what the **andrews1993tests<empty citation>** critical values correct for, and why we use the test only to reject constancy.

Interpretation. The two objects agree, and they support a claim about a period rather than a date: strategic silence is weak and statistically absent through 2020 and intensifies sharply over 2021–2023. The monotone Wald surface is exactly what this trajectory implies mechanically – pushing the cutoff later isolates a progressively purer post-period of high-severity years, so the pre/post contrast keeps growing – and it is why the scan cannot separate a late break from a late-concentrated trend. The intensification coincides with a broad shift in climate scrutiny: the surge in sustainable investment, U.S. re-engagement with Paris, and the build-up to the SEC’s proposed climate-disclosure rule. We do *not* attribute it to any single rule. The annual test cannot resolve a specific date within 2021–2023, and it is silent on the March 2022 SEC proposal, which we examine as a separate, higher-frequency test below.

5.5.2 SEC event study

- Event: SEC climate-disclosure proposal, March 21, 2022. Sample: never-disclosing firms. MNAR emissions measured in 2021. Sector FEs; heteroskedasticity-robust SEs.
- CARs regressed on log MNAR emissions:
 - $[-1, +1]$ CAR: -0.0041^{***}
 - $[-5, +5]$ CAR: -0.0078^{***}
 - $[-30, +30]$ CAR: $+0.0025$ (insig)
- **MNAR placebo passes cleanly** on the same calendar date in 2019: all three windows near-zero and insignificant.
- **Trucost placebo fails:** on the placebo date, Trucost shows significant wrong-signed coefficients ($[-1, +1]$ $+0.0015^*$, $[-30, +30]$ -0.0152^{***}) while MNAR remains insignificant – indicating identification contamination in vendor data and cleaner identification from the MNAR estimator.
- Horse race on the SEC date: MNAR retains an independent $[-5, +5]$ effect (-0.0061^{**}) in a combined specification.
- MAR vs. MNAR comparison: nearly identical on both dates; the exponential tilt adds limited additional information for this test.

[TODO: Re-verify multi-date placebo claim; current evidence is on 2019 only.]

5.5.3 Transition-firm DiD: asymmetric repricing

- Sample: transition firms (non-disclosure $\rightarrow$ voluntary disclosure). High Emitter = top-quartile disclosed emissions at $t = 0$ (ground truth); MNAR-imputed emissions at $t - 1$ provide a robustness check.
- Three event windows: $[-2, +2]$, $[-2, +4]$, $[-4, +4]$.
- Pooled High $\times$ Post coefficient:
 - $[-2, +2]$: $+0.651^{**}$
 - $[-2, +4]$: $+0.561^{**}$
 - $[-4, +4]$: $+0.537^{**}$

- Post-disclosure level effect negative and significant (-0.382^{***} , -0.376^{***} , -0.283^{**}): the average transition firm sees lower returns post-disclosure, but high emitters are repriced *upward* relative to low emitters.
- Pre-period High Emitter coefficient: statistically zero across windows (-0.270 , -0.303 , -0.239 , all insig). Parallel trends pass.
- Placebo with disclosure shifted back three years: $\text{High} \times \text{Post} = +0.044$ (clean placebo).

[TODO: MNAR-imputed-treatment robustness and median-emitter robustness: rerun if reported.]

Cohort decomposition.

- 2013– 2016 cohort: $+0.703$ (insig).
- 2017– 2019 cohort: -0.655 (insig).
- **2020– 2023 cohort:** $+1.217^{***}$.

The pattern is consistent across windows: the 2020– 2023 cohort is $+1.097^{**}$ at $[-2, +4]$ and $+0.958^{***}$ at $[-4, +4]$; the two earlier cohorts are insignificant in every window.

Interpretation of the regime evidence. The three pieces of evidence above are mutually independent – a year-by-year propensity-model parameter, a daily-frequency event study, and a monthly cohort-DiD – and we read them in parallel rather than chaining them causally. Together they place the intensification of carbon-disclosure relevance in the early 2020s: strategic silence deepens in the propensity model from around 2020, the market reprices recovered emissions around the March 2022 SEC proposal, and high-emission firms that newly disclose are repriced upward in the 2020– 2023 cohort. Voluntary disclosure in this period functions as a credibility/bonding signal in the shadow of prospective mandatory regimes: a high-emission firm that discloses despite the regulatory cost reveals commitment that the market rewards.

6 Aggregate Implications

6.1 U.S. total Scope 1 emissions

The MNAR correction allows us to estimate aggregate Scope 1 emissions for the universe of U.S. public firms listed on the major exchanges – including the 87% of firm-years that never voluntarily disclose. [TODO: Insert 2022 estimate in MtCO₂; tilt adjustment as share of missing-firm total.]

6.2 Corporate carbon damages

Applying the social cost of carbon framework of Greenstone et al. (2023) with mid-scenario SCC of \$190/tCO₂, our damages are 2.5– 3× comparable benchmarks for the disclosed sample. The implication is that voluntary disclosure regimes systematically understate corporate carbon damages, and the understatement concentrates in the firms with the strongest strategic incentive to remain silent.

[TODO: One headline table; one paragraph. This is a downstream implication, not a core contribution.]

7 Conclusion

This paper makes three contributions to the joint literatures on voluntary disclosure theory and the asset pricing of carbon risk. First, we provide a game-theoretic microfoundation for missing-not-at-random selection in corporate emissions disclosure. The equilibrium logistic propensity and the exponential-tilting correction emerge as objects of the disclosure game, not as imposed modeling assumptions. A single structural parameter – the selection severity index γ – governs the strength of strategic silence, and its cross-sectional and temporal variation map directly onto observable empirical predictions.

Second, equity markets price the act of disclosure itself. A transparency premium of +0.225*** pp/month rewards voluntary disclosure across every sample specification we examine, orthogonal to the level of emissions disclosed. This is the empirical analog of the classical Milgrom– Verrecchia logic: voluntary disclosure is rewarded because silence is penalized.

Third, and most importantly, strategic silence operates exactly where voluntary disclosure theory predicts an audience exists. MNAR selection concentrates in high-regulatory sectors and high-institutional-ownership firms, and reverses in low-attention contexts. The Milgrom– Verrecchia mechanism is not a universal feature of disclosure environments; it is a conditional one, and the conditions are observable. The cross-sectional heterogeneity in $\hat{\gamma}_{\text{post}}$ is the theory’s empirical fingerprint, not a complication for it.

The mechanism intensifies in a regulatory-anticipation regime over the early 2020s, on which three independent pieces of evidence bear: a sharp rise in $\hat{\gamma}_t$ over 2021–2023 after a decade in which it is statistically indistinguishable from zero, an SEC event-study response among high-MNAR non-disclosers, and a concentrated post-disclosure repricing of high-emission firms in the 2020– 2023 cohort. We date this to a period rather than a year: a cutoff-sensitivity scan rejects constancy of γ but cannot locate a break, and we do not claim one. The asymmetric repricing of high-emission transition firms during this regime suggests that the prospect of mandatory disclosure already affects equity pricing through firms’ voluntary-disclosure choices, before mandatory rules are finalized.

Beyond pricing, 87% of U.S. public firm-years on the major exchanges carry no voluntarily disclosed Scope 1 figure. Voluntary disclosure regimes therefore systematically understate the corporate emissions distribution. MNAR correction raises aggregate corporate Scope 1 emissions by $\sim 12\%$ and corporate carbon damages by $2.5\text{--}3\times$ relative to disclosed-sample benchmarks. The understatement is concentrated in the firms with the strongest strategic incentive to remain silent – precisely the firms whose emissions matter most for both asset pricing and climate policy.

Policy implication. Mandatory disclosure regimes would extend carbon-risk pricing to the bulk of the public corporate sector that voluntary frameworks miss. The market is already partially performing this function in advance of mandatory rules, by pricing the act of disclosure itself; mandatory regimes would shift this signal from a chosen one to a uniform one and remove the strategic-silence margin entirely.

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A Proofs of Theoretical Results

Throughout, $\Lambda(z) = (1 + e^{-z})^{-1}$ denotes the standard logistic CDF, which satisfies $1 - \Lambda(z) = \Lambda(-z)$ and $\frac{1 - \Lambda(z)}{\Lambda(z)} = e^{-z}$.

A.1 Proof of Lemma ??.

If firm i discloses, $\Omega = \{y_i\}$, so $\mathbb{E}[y \mid \Omega] = y_i$ and $P_D(y_i) = V_0 - \beta y_i$. If it withholds, $\Omega = \{\delta_i = 0\}$; because firms form a continuum, the report of the measure-zero firm i does not move the pool, so $\mathbb{E}[y \mid \Omega] = \mu_{ND}$ independently of y_i and $P_{ND} = V_0 - \beta \mu_{ND}$. Firm i discloses iff $P_D(y_i) - \tilde{c}_i \geq P_{ND}$, i.e.

$$V_0 - \beta y_i - c_0(x_i) - c_1 y_i - \lambda \tilde{\eta}_i \geq V_0 - \beta \mu_{ND}.$$

Rearranging and isolating $\tilde{\eta}_i$ gives $\tilde{\eta}_i \leq f(x_i) - \gamma y_i$ with $f(x_i) = \frac{1}{\lambda}[\beta \mu_{ND} - c_0(x_i)]$ and $\gamma = (\beta + c_1)/\lambda$, which is (??). Since $\tilde{\eta}_i$ is standard logistic and independent of (y_i, x_i) , $\mathbb{P}(\delta_i = 1 \mid x_i, y_i) = \mathbb{P}(\tilde{\eta}_i \leq f(x_i) - \gamma y_i) = \Lambda(f(x_i) - \gamma y_i)$. $\square$

A.2 Proof of Theorem ??

Given a candidate pooling mean m , Lemma ?? gives the disclosure rule $\Lambda(f(x_i; m) - \gamma y_i)$ with $f(x_i; m) = \frac{1}{\lambda}[\beta m - c_0(x_i)]$, which in turn induces a pooling mean

$$T(m) = \mathbb{E}[y \mid \tilde{\eta} > f(x; m) - \gamma y] = \frac{\mathbb{E}[y (1 - \Lambda(f(x; m) - \gamma y))]}{\mathbb{E}[1 - \Lambda(f(x; m) - \gamma y)]}.$$

An equilibrium is a fixed point $m^* = T(m^*)$, with $f^* \equiv f(\cdot; m^*)$. Since y has finite mean under the prior and $1 - \Lambda \in (0, 1)$, T is continuous and maps the convex hull of the support of y into itself, so a fixed point exists by Brouwer's theorem. [TODO: Uniqueness: T is a contraction under a bound on β/λ ; and $\mathbb{E}[e^{\gamma y} \mid \delta = 1] < \infty$, which holds since y is Gaussian under the prior.] The slope $\gamma = (\beta + c_1)/\lambda$ in (??) does not depend on m ; multiplicity of f^* , were it to arise, would relocate the intercept but leave γ unchanged. $\square$

A.3 Proof of Corollary ??

For $\delta \in \{0, 1\}$, Bayes' rule gives $f_\delta(y | x) = \frac{\mathbb{P}(\delta | x, y) f(y | x)}{\mathbb{P}(\delta | x)}$, where $f(y | x)$ is the population emission density. Taking the ratio of the two densities and using (??),

$$\frac{f_0(y | x)}{f_1(y | x)} = \frac{\mathbb{P}(\delta = 0 | x, y)}{\mathbb{P}(\delta = 1 | x, y)} \cdot \frac{\mathbb{P}(\delta = 1 | x)}{\mathbb{P}(\delta = 0 | x)} = \frac{1 - \Lambda(f^* - \gamma y)}{\Lambda(f^* - \gamma y)} C(x) = e^{\gamma y - f^*} C(x),$$

where $C(x) = \mathbb{P}(\delta = 1 | x) / \mathbb{P}(\delta = 0 | x)$ is independent of y . Hence $f_0(y | x) \propto e^{\gamma y} f_1(y | x)$, and normalizing so that $\int f_0 dy = 1$ fixes the constant at $1 / \mathbb{E}[e^{\gamma \tilde{y}} | x, \delta = 1]$, giving (??). $\square$

A.4 Proof of Proposition ??

Let $K(\gamma) \equiv \log \mathbb{E}[e^{\gamma \tilde{y}} | \delta = 1]$ be the cumulant generating function of the disclosing-firm emission density f_1 (finite for all γ since y is Gaussian under the prior). By Corollary ??, f_0 is the γ -exponential tilt of f_1 , so

$$\mathbb{E}[y | \delta = 0] = \frac{\mathbb{E}[y e^{\gamma y} | \delta = 1]}{\mathbb{E}[e^{\gamma y} | \delta = 1]} = K'(\gamma), \quad \mathbb{E}[y | \delta = 1] = K'(0).$$

Therefore $\Delta\mu(\gamma) = K'(\gamma) - K'(0)$ and $\frac{d\Delta\mu}{d\gamma} = K''(\gamma) = \text{Var}_\gamma[y] > 0$, where Var_γ is the variance under the tilted law f_0 , strictly positive because emissions are non-degenerate. Integrating from $\Delta\mu(0) = 0$ yields $\Delta\mu(\gamma) > 0$ for all $\gamma > 0$. $\square$

A.5 Proof of Proposition ??

With the transparency benefit $b y_i$ added to the disclosure payoff, firm i discloses iff $P_D(y_i) + b y_i - \tilde{c}_i \geq P_{ND}$, i.e.

$$V_0 - \beta y_i + b y_i - c_0(x_i) - c_1 y_i - \lambda \tilde{\eta}_i \geq V_0 - \beta \mu_{ND}.$$

Isolating $\tilde{\eta}_i$ as in the proof of Lemma ?? gives $\tilde{\eta}_i \leq f(x_i) - \gamma_{\text{eff}} y_i$ with $\gamma_{\text{eff}} = (\beta + c_1 - b) / \lambda$. The sign conditions follow immediately: $\gamma_{\text{eff}} > 0 \iff b < \beta + c_1$ and $\gamma_{\text{eff}} < 0 \iff b > \beta + c_1$. $\square$

B Estimation and Imputation Details

This appendix details the estimation steps summarized in Section ??: the outcome model and baseline imputation, the construction of the excluded shifter, the disclosure (response) mechanism, the exponential-tilt adjustment, and the Monte Carlo propagation of imputation uncertainty. Figure ?? provides an overview of the procedure.

B.1 Outcome model and baseline imputation

The first step estimates the conditional distribution of emissions among disclosing firms. We fit the lognormal outcome model (??) by ordinary least squares on the subsample of disclosers, with covariates comprising log total assets, GICS industry fixed effects, year fixed effects, and the oil-price exposure shifter z_{it} defined below. Total assets proxy production scale while avoiding the valuation-driven volatility of market capitalization; the industry and year effects absorb, respectively, the cross-industry emissions gradient and common time variation. The fit yields $\hat{\eta}$ and the residual variance $\hat{\sigma}^2$, which characterize the conditional density $f_1(y | \mathbf{x})$ of disclosed emissions.

A useful way to think about this first step is as recovering the production-side signal in emissions that is observable among firms that choose to disclose. Conditional on disclosure, Scope 1 emissions are largely driven by scale, industry technology, and aggregate economic conditions. Equation (??) formalizes this intuition by mapping emissions to observables that proxy for these underlying determinants, while treating residual variation as idiosyncratic noise.

Importantly, Equation (??) is estimated on the subsample of disclosing firms, to obtain the consistent estimator $\hat{\eta} = (\hat{\eta}_0, \hat{\eta}_1)$ that characterizes the conditional distribution of emissions $f_1(y | \mathbf{x}, \hat{\eta})$. This step does not attempt to model disclosure behavior itself; rather, it isolates the conditional emissions process that would prevail absent selection. Under MAR, this model would suffice to impute missing values.

In the presence of non-ignorable disclosure, this step serves as a baseline to generate initial imputations for non-reporting firms. Specifically, we draw from the predictive distribution

$$\hat{y}_{it} | \mathbf{x}_{it} \sim \mathcal{N}(\hat{\eta}_0 + \mathbf{x}_{it}^\top \hat{\eta}, \hat{\sigma}^2). \quad (14)$$

This step generates counterfactual emissions under the assumption that non-reporting firms follow the same conditional distribution as reporting firms, i.e., $f_1(y|\mathbf{x}, \hat{\eta}) = f_0(y|\mathbf{x}, \hat{\eta})$. As previously discussed, this assumption is generally violated when disclosure is endogenous. The subsequent steps adjust these imputations to account for selection.

B.2 The excluded shifter

Identification of the emissions slope requires a shifter of emissions that is excluded from the disclosure decision (Wang et al., 2014). Partitioning the covariates as $\mathbf{x}_{it} = (\mathbf{x}_{1,it}, z_{it})$, where $\mathbf{x}_{1,it}$ (firm size and sector) enters the disclosure rule, identification requires that, conditional on $\mathbf{x}_{1,it}$ and y_{it} , the shifter does not affect disclosure:

$$\mathbb{P}(\delta_{it} = 1 \mid \mathbf{x}_{it}, y_{it}) = \mathbb{P}(\delta_{it} = 1 \mid \mathbf{x}_{1,it}, y_{it}). \quad (15)$$

We construct the shifter in shift-share (Bartik) form (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020),

$$z_{it} = \log(\text{WTI}_t) \times \text{exposure}_i. \quad (16)$$

The exposure is firm i 's *predetermined* physical-capital intensity – gross property, plant, and equipment scaled by total assets, averaged over the firm's first (up to) five available observations and held fixed thereafter. To keep the interaction well-behaved we winsorize the exposure at its 1st and 99th percentiles and standardize it. Firms with no reported gross PP&E in any of those years are dropped rather than imputed; as reported in Table ??, this removes 7,441 firm-years, overwhelmingly Financials (6,031) and Real Estate (1,647). We prefer dropping to assigning these firms a sector-median exposure because the fuel-combustion channel the instrument relies on does not operate for them: a fabricated exposure would enter z_{it} as noise interacted with the oil price, weakening relevance without adding identifying variation. The cost is that the recovered sample excludes those two sectors, which is a limitation we state plainly rather than paper over; it is also why Section ?? does not report a sectoral $\hat{\gamma}$ for them. Because plain WTI_t is a year-level series, it is collinear with the year fixed effects and contributes no independent variation; the *interaction* z_{it} varies across firms within a year and is the object that supplies identifying variation, present in the emissions equation but excluded from the disclosure equation. Exclusion rests on the exogeneity

of the aggregate shock (Borusyak et al., 2022): the crude oil price is exogenous to any individual firm’s disclosure choice, and firms differ only in predetermined exposure. Appendix ?? reports the estimates obtained when the WTI shock is replaced by U.S. real GDP.

B.3 Response mechanism

We model the probability of emissions disclosure using a response mechanism that depends on firm characteristics and emissions, restating (??) for convenience:

$$\mathbb{P}(\delta_i = 1 \mid \mathbf{x}_{1i}, y_i) = \frac{\exp\{f(\mathbf{x}_{1i}) + g(y_i)\}}{1 + \exp\{f(\mathbf{x}_{1i}) + g(y_i)\}}, \quad (17)$$

where δ_i is the disclosure indicator, $\mathbf{x}_{1i}$ includes firm size and sector, and $g(y_i)$ captures the dependence of disclosure on emissions.

We estimate the disclosure model (??) in two forms. In the fully parametric (FP) specification – our primary specification – $f(\mathbf{x}_{1,it})$ is linear in log assets and sector fixed effects and $g(y_{it}) = -\gamma y_{it}$ is linear in emissions, so the model is an ordinary logistic regression whose coefficient on y_{it} estimates $-\hat{\gamma}$; this is the form the disclosure game implies (Lemma ??). In the semiparametric (SP) specification, f and g are generalized additive models with penalized splines on log assets and log emissions, allowing the disclosure response to vary nonlinearly (Yang & Kim, 2017). Comparing the two isolates the role of the linearity assumption; both are reported in Section ?? and Appendix ??.

B.4 Exponential-tilt adjustment

Given the estimated response mechanism, we correct the baseline imputations for selection using the exponential tilt of Corollary ??. Under the logistic specification, the density of unobserved emissions differs from the observed density by a factor proportional to $\exp\{-g(y)\}$, which in the fully parametric case is $\exp\{\hat{\gamma} y\}$. We therefore reweight each imputation by $w_{it} \propto \exp\{\hat{\gamma} \hat{y}_{it}\}$ (equation (??)), so that firms with lower disclosure probabilities – the higher emitters – receive greater weight, restoring the upper-tail mass that selection removes. Weights are trimmed at the 1st and 99th percentiles to limit the influence of extreme draws.

Intuitively, firms with emissions levels associated with a lower probability of disclosure

receive greater weight in the adjusted distribution. This procedure restores right-tail mass that is systematically under-represented among reporting firms and yields selection-corrected estimates of firm-level and aggregate emissions.

B.5 Monte Carlo sensitivity

We quantify imputation uncertainty by repeating the imputation–tilt procedure across Monte Carlo draws. For each draw $b = 1, \dots, B$: (i) sample baseline imputations $\{\hat{y}_{it}^{(b)}\}$ for the non-disclosers from (??); (ii) re-estimate the response mechanism (??) on the completed data; and (iii) form the tilting weights (??) and compute the selection-corrected quantities. We report the mean and standard error across $B \in \{300, 500\}$ draws (Gohil, 2020). Because the missing-not-at-random restriction is not testable, these draws characterize sampling and specification variability rather than provide a formal test (Little & Rubin, 2019).

B.6 Robustness

We probe the primary specification along four dimensions: the aggregate shock entering the excluded shifter, the granularity of the fixed effects, the functional form of the disclosure model, and the number of Monte Carlo draws. Each varies one element and holds the rest fixed. We also report the two identification diagnostics that motivate sample and reporting choices made in the body: the degeneracy of the disclosure margin off the major exchanges, and the sector-level disclosure rates behind the 10% reporting threshold of Section ???. Three of the four checks leave the correction essentially unchanged; the exception is fixed-effect granularity, where the tilt is sensitive in a direction we explain below and which makes our headline estimate the conservative one.

Where γ is not identified. Two of our design choices amount to the same rule – do not report a selection severity where disclosure does not vary – and we document the evidence for both here.

Table ?? rebuilds the panel without the exchange filter of Section ?? and reports the disclosure margin by listing venue. The asymmetry is extreme. Among major-exchange firm-years, 11.7% carry a disclosed Scope 1 figure. Among the 21,714 off-exchange firm-years – OTC, pink-sheet, and foreign venues – *seven* do: a disclosure

rate of 0.03%. Estimating an emissions slope on that group anyway returns $|\hat{\gamma}| \approx 13$, an odds ratio of e^{13} per standard deviation, which is not an economic magnitude but the numerical signature of a quasi-separated logit fit to seven positive outcomes.

It is worth being precise about what this does and does not establish, because the natural argument is not the correct one. These firms do *not* distort the pooled estimate: the pooled slope (-0.019) is indistinguishable from the major-exchange slope (-0.018), because a group with seven disclosers contributes almost nothing to the likelihood of the emissions coefficient. The case for excluding them is the converse, and it is stronger. They cannot inform γ at all, yet retaining them would assign 19,097 firm-years an imputed emissions level and an exponential tilt whose selection severity *and* outcome model are extrapolated wholesale from firms with whom they share no disclosure behavior. We would be recovering emissions for a population about whose disclosure decision the data are silent, and reporting the result as though it were measured. The restriction is therefore a statement about the reach of the estimator, not a claim that the excluded firms contaminate it.

The same pathology appears by sector, where Financials and Real Estate – which disclose rarely and, per Appendix ??, lack the physical-capital exposure the instrument requires – produce estimates an order of magnitude larger than any identified sector, with intervals to match. [TODO: Insert the table of disclosure rate by sector alongside each sector’s unrestricted $\hat{\gamma}$, so the reader can see exactly what the 10% threshold of Section ?? removes.]

Alternative shifters. The shift-share instrument combines an aggregate shock with a predetermined firm-level exposure. We vary the shock, replacing $\log(\text{WTI}_t)$ with $\log$ U.S. real GDP while holding the exposure and every other element of the specification fixed, so that the shock is the only thing that changes (Table ??). The substitution leaves the results essentially untouched: the shifter remains strongly relevant in the outcome model ($\hat{\eta}_2 = 0.348$, s.e. 0.011, against 0.188 under WTI), the fit is identical to three decimals ($R^2 = 0.809$ under both, on the same 6,698 disclosers), the size elasticity is unchanged (0.970 versus 0.968), and the tilt correction moves from +11.9% to +11.5%.

This near-equivalence is a structural feature of the estimator rather than a coincidence, and it is worth stating explicitly because two nearly identical columns invite the suspicion that the instrument does no work. The selection severity is identified from the propensity model, which *excludes* the shifter by construction; the shifter enters only

through the outcome model, where it shapes the imputed emissions of non-disclosers. Two shocks that shift emissions comparably therefore deliver comparable $\hat{\gamma}$, and the correct reading of the stability is the reassuring one: $\hat{\gamma}$ is not an artifact of which aggregate series we interact with exposure. What the exercise does *not* test is the exposure side of the shift-share, which is where the identifying within-year variation actually comes from; we take that up next.

Fixed-effect granularity. The baseline uses GICS industry fixed effects. Re-estimating with the coarser GICS-11 sector classification, holding the WTI shifter and the sample fixed (Table ??), leaves the qualitative structure intact: the size elasticity remains near one (1.027 versus 0.968), the shifter remains strongly relevant ($\hat{\eta}_2 = 0.271$), and the brown/green ordering is preserved, with Utilities (+3.79), Materials (+3.48), and Energy (+3.14) at the top and Financials (−1.70) at the bottom. The fit falls as expected with fewer effects ($R^2 = 0.738$ versus 0.809).

The correction, however, is *not* invariant to this choice, and we report the sensitivity rather than smooth it over. Under sector fixed effects the tilt rises to +21.6%, against +11.9% under industry effects – close to double. The mechanism is transparent and, in our reading, favors the baseline. Coarser fixed effects leave more genuine cross-firm heterogeneity in emissions unexplained, inflating the residual variance σ^2 of the outcome model; since the Gaussian bias in (??) scales as $\Delta\mu = \gamma\sigma_1^2$, a fatter residual distribution mechanically produces a larger tilt for the same selection severity. The sector specification is thus attributing to selection what is in fact unmodeled industry technology. This is precisely why we adopt the finer industry fixed effects as the baseline: they absorb the production-side variation that the tilt would otherwise misread as strategic silence, and they make the reported correction the *conservative* of the two. A reader who prefers the coarser specification should read our headline figure as a lower bound.

Functional form. We compare the fully parametric (FP) disclosure model with a semiparametric generalized additive model (SP) – specifically, a Logistic Generalized Additive Model (LogisticGAM). This approach introduces multiple spline regressors, allowing for greater flexibility in capturing non-linear relationships between the covariates and the probability of disclosure, and without requiring any specification of the response model. However, this increased flexibility comes at the cost of interpretability, as the estimated coefficients vary across the range of the covariates.

Figure ?? presents the estimated standardized coefficient values for firm size, Scope 1

Table A1: **Outcome model, coarser fixed effects (robustness).** Linear regression of log Scope 1 emissions on firm characteristics, the shift-share oil shifter, GICS-11 *sector* indicators, and year fixed effects. Identical to the baseline of Table ?? except that the finer GICS industry effects are replaced by sector effects; sample, shifter, and all other choices are held fixed. Robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Log Scope 1 Emissions
<i>Macroeconomic and firm characteristics</i>	
z_{it} (oil-price shifter)	0.2706*** (0.006)
$\log(\text{Assets}_i)$	1.0271*** (0.014)
Constant	0.2258 (0.206)
<i>Sector fixed effects (baseline: Communication Services)</i>	
Consumer Discr	1.3687***
Consumer Staples	2.7184***
Energy	3.1446***
Financials	-1.6971***
Health Care	1.0760***
IT	0.5986***
Industrials	2.3802***
Materials	3.4809***
Real Estate	1.2097***
Utilities	3.7898***
<i>Year fixed effects (baseline: 2012)</i>	
2013	-0.0397
2014	-0.0832
2015	-0.0705
2016	-0.1043
2017	-0.2100*
2018	-0.2606**
2019	-0.3124***
2020	-0.3439***
2021	-0.5152***
2022	-0.5612***
2023	-0.7031***
Observations	6,698
R^2	0.738

Table A2: **Outcome model, alternative shock (robustness)**. Linear regression of log Scope 1 emissions on firm characteristics, the shift-share shifter built from log U.S. real GDP in place of the WTI crude price, GICS industry indicators, and year fixed effects. Identical to the baseline of Table ?? except for the aggregate shock; the predetermined firm exposure, the sample, and all other choices are held fixed. Robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Log Scope 1 Emissions
<i>Macroeconomic and firm characteristics</i>	
z_{it} (GDP shifter)	0.3476*** (0.011)
$\log(\text{Assets}_i)$	0.9697*** (0.013)
Constant	3.8844*** (0.183)
<i>Year fixed effects (baseline: 2012)</i>	
2013	-0.0802
2014	-0.1244
2015	-0.1620
2016	-0.2277**
2017	-0.3136***
2018	-0.3287***
2019	-0.3791***
2020	-0.4537***
2021	-0.5541***
2022	-0.5822***
2023	-0.7101***
<i>Industry fixed effects (baseline: Energy Equipment & Services)</i>	
Observations	6,698
R^2	0.809



Figure A1: **Semiparametric response mechanism.** Distribution of standardized coefficients from the GAM disclosure model across $B = 300$ Monte Carlo draws: firm size (left), Scope 1 emissions (centre), and sector indicators (right). Red lines mark the mean across draws; bands and whiskers show variability across draws. The emissions-disclosure relationship is approximately linear and negative, supporting the linear specification the disclosure game implies.

emissions, and the sector covariates, on the probability of emissions disclosure, along with their respective 95% confidence intervals. The left panel illustrates a non-linear relationship between firm size and disclosure. Specifically, the effect increases rapidly for small- to mid-sized firms and then stabilizes, indicating diminishing marginal effects at higher asset levels. Interestingly, firms in the lowest asset quantile exhibit a negative coefficient, suggesting that very small companies are less likely to disclose, potentially due to the fixed costs of reporting and limited regulatory and stakeholder pressure.

The middle panel highlights a clear, approximately linear negative relationship between Scope 1 emissions and disclosure likelihood. As emissions increase, the associated coefficient steadily declines, implying that firms with higher emissions are systematically less likely to disclose. This pattern is consistent with non-random disclosure, in which higher emitters are less likely to disclose, aligning with the MNAR framework adopted in the analysis.

The right panel displays the estimated sectoral effects relative to the baseline (Communication Services). Results align with fully parametric ones. Most sectors exhibit positive coefficients, indicating a higher probability of disclosure compared to the baseline.

Despite the advantages of the semi-parametric approach in capturing complex relationships, its interpretability remains limited due to the non-parametric nature of the splines. Nonetheless, the trends observed align with expectations and provide a nuanced understanding of the covariates' influence on disclosure probability, while validating the robustness of the fully parametric results. The tilt correction is larger under the SP model (about +19.1% versus +12.4% under FP at $B = 500$; +19.4% versus +11.9% at $B = 300$; Table ??), reflecting the additional upper-tail flexibility of the splines. We adopt FP as primary because it is the specification the disclosure game implies, and read the SP figure as an upper bound on the correction: the two bracket the plausible range, and the qualitative conclusion – that MAR imputation understates emissions materially, and understates them in the tail – holds under either. The gap between them is itself informative. It says the linear $g(y)$ is mildly conservative relative to a flexible fit, which is the direction of error we would rather make, since the tilt compounds any overstatement of $\hat{\gamma}$ into the aggregate.

Monte Carlo draws. All Monte Carlo quantities are reported for both $B = 300$ and $B = 500$ draws (Tables ?? and ??). The means are essentially identical across the two, indicating that the procedure has converged and that 300 draws suffice.

C Tables

C.1 Sample Statistics

Table A3: **Sample construction.** Firm-year observations retained at each step, from the raw Compustat North America fundamentals extract to the estimation sample. The exchange restriction and the predetermined-exposure requirement are discussed in the text.

Step	Firm-years	Dropped
Compustat North America fundamentals	161,717	
Non-missing identifiers and core financials	145,805	15,912
Major U.S. exchanges (NYSE, NYSE American, Nasdaq)	89,637	56,168
Merged to Trucost Environmental and macro series	71,261	18,376
Valid disclosure classification and total assets	69,746	1,515
Sample window 2012–2023	59,996	9,750
Total assets trimmed at 1st/99th percentile within year	58,796	1,200
Predetermined PP&E exposure available	51,355	7,441
Estimation sample	51,355	
<i>Unique firms</i>	6,938	
<i>Average firms per year</i>	4,280	
<i>Share with disclosed Scope 1</i>	13.0%	

⁸CO₂ Emission divided by total assets of a company (CO₂ Metric Tonnes per \$Million).

Table A4: **Sample Summary Statistics.** Data are extracted from Trucost Environmental and Compustat, for US public companies from 2012 to 2023. Data are yearly values. Total Market Value is in trillion USD. Total CO₂ Emissions are Scope 1 Emissions, measured in billions of metric tons (Gt), both disclosed by companies and estimated by Trucost. Average CO₂ intensity is computed as total emissions per total assets of each company, in Tons of CO₂ over Millions USD.

Year	N	Total Mkt Value	Scope 1 Emissions	Intensity ⁸
2012	4739	15.21	2.18	287.81
2013	4821	19.13	2.22	279.53
2014	4786	21.22	2.23	266.77
2015	4739	19.99	2.10	261.67
2016	4716	21.18	2.20	166.12
2017	4717	25.22	2.35	164.06
2018	4727	23.30	2.45	157.98
2019	4862	26.32	2.33	164.13
2020	5014	30.47	2.13	162.95
2021	5149	38.16	2.32	150.84
2022	5232	30.13	2.19	109.79
2023	5294	35.70	2.06	104.63

Table A5: **Summary statistics by year and disclosure status.** Firms are split by whether Scope 1 emissions are directly disclosed (*Observed*) or are absent or vendor-estimated (*Missing*); see Table ?? for the classification. Total market value is in trillion USD; Scope 1 emissions in billions of metric tons (Gt); intensity is emissions per total assets (tCO₂ per \$Million). Figures describe the sample after the asset trim and before the predetermined-exposure requirement (58,796 firm-years; see Table ??).

Year	N		Total Mkt Value		Scope 1 Emissions		Intensity	
	Missing	Obs.	Missing	Obs.	Missing	Obs.	Missing	Obs.
2012	4,434	305	7.27	7.95	0.39	1.79	151.85	464.34
2013	4,501	320	9.21	9.92	0.27	1.95	148.96	470.90
2014	4,452	334	9.98	11.25	0.20	2.03	125.16	464.78
2015	4,402	337	9.48	10.50	0.31	1.79	142.25	433.18
2016	4,335	381	9.37	11.81	0.41	1.80	107.47	420.27
2017	4,294	423	11.06	14.16	0.42	1.93	106.25	400.22
2018	4,243	484	9.14	14.15	0.46	1.99	97.08	378.45
2019	4,298	564	10.00	16.31	0.25	2.09	82.75	414.34
2020	4,287	727	11.49	18.98	0.19	1.94	68.02	413.66
2021	4,231	918	10.44	27.73	0.21	2.11	58.48	336.16
2022	4,107	1,125	6.34	23.79	0.11	2.07	43.11	224.00
2023	4,320	974	9.51	26.19	0.25	1.81	51.08	219.44

Table A6: **Sources of Scope 1 emissions data, 2023 snapshot.**

Disclosure source	Count	Share (%)	Treatment
<i>Vendor-estimated emissions</i>			
Estimated data	1,894	35.78	Missing
Derived from previous year	173	3.27	Missing
Value derived from fuel use provided in Environmental/CSR	12	0.23	Missing
Value derived from fuel use provided in Annual Report/Financial Accounts Disclosure	6	0.11	Missing
Estimate based on partial data disclosure in Environmental/CSR	2	0.04	Missing
Estimate based on partial data disclosure in Annual Report/10-K/Financial Accounts	1	0.02	Missing
<i>Firm-reported emissions (CSR / Environmental reports)</i>			
Value derived from data provided in Environmental/CSR	724	13.68	Observed
Exact Value from Environmental/CSR	104	1.96	Observed
Value summed up from data provided in Environmental/CSR	29	0.55	Observed
Value split from data provided in Environmental/CSR	9	0.17	Observed
<i>Firm-reported emissions (CDP)</i>			
Exact Value from CDP	52	0.98	Observed
Value derived from data provided in CDP	30	0.57	Observed
<i>Firm-reported emissions (Annual reports / 10-Ks)</i>			
Value derived from data provided in Annual Report/Financial Accounts Disclosure	22	0.42	Observed
Exact Value from Annual Report/10K/Financial Accounts Disclosure	2	0.04	Observed
Value summed up from data provided in Annual Report/Financial Accounts Disclosure	1	0.02	Observed
Value split from data provided in Annual Report/Financial Accounts Disclosure	1	0.02	Observed
<i>Missing</i>			
None	2,232	42.16	Missing
Total	5,294	100.00	

C.2 Outcome Model Results

Table A7: **Outcome model results.** Linear regression of log Scope 1 emissions on firm characteristics, macroeconomic conditions, industry indicators, and year fixed effects. Robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Log Scope 1 Emissions
<i>Macroeconomic and firm characteristics</i>	
Oil exposure $\times$ log(WTI _{<i>t</i>})	0.1883*** (0.006)
log(Assets _{<i>i</i>})	0.9675*** (0.013)
Constant	3.8807*** (0.183)
<i>Year fixed effects (baseline: 2012)</i>	
2013	-0.0851
2014	-0.1237
2015	-0.1097
2016	-0.1666
2017	-0.2634***
2018	-0.2994***
2019	-0.3385***
2020	-0.3849***
2021	-0.5284***
2022	-0.5748***
2023	-0.6915***
<i>Industry fixed effects (baseline: Energy Equipment & Services)</i>	
Observations	6,698
R^2	0.809

C.3 Response Mechanism Results

Table A8: **Response mechanism: fully parametric model.** Mean coefficient estimates and standard errors from Monte Carlo simulations. Results are reported for $B = 300$ and $B = 500$ draws.

	$B = 300$		$B = 500$	
	Mean	SE	Mean	SE
<i>Firm characteristics</i>				
Constant	-2.9154	0.0112	-2.9165	0.0118
Size	1.8701	0.0265	1.8722	0.0275
Emissions	-0.1953	0.0727	-0.1994	0.0743
<i>Sector effects (baseline: Communication Services)</i>				
Consumer Discretionary	1.4806	0.0067	1.4812	0.0069
Consumer Staples	2.0418	0.0063	2.0423	0.0065
Energy	1.4992	0.0171	1.5004	0.0177
Financials	-0.3936	0.0026	-0.3938	0.0027
Health Care	0.1659	0.0057	0.1663	0.0059
IT	1.3505	0.0076	1.3511	0.0078
Industrials	1.8437	0.0108	1.8445	0.0111
Materials	2.3713	0.0185	2.3725	0.0191
Real Estate	-0.0703	0.0066	-0.0698	0.0068
Utilities	1.9695	0.0728	1.9741	0.0751
<i>Year fixed effects (baseline: 2012)</i>				
2013	0.0903	0.0044	0.0907	0.0048
2014	0.0619	0.0042	0.0618	0.0041
2015	0.1035	0.0042	0.1034	0.0043
2016	0.3242	0.0045	0.3242	0.0044
2017	0.4613	0.0049	0.4611	0.0049
2018	0.6256	0.0050	0.6255	0.0049
2019	0.8757	0.0043	0.8757	0.0044
2020	1.1888	0.0049	1.1888	0.0049
2021	1.4502	0.0046	1.4502	0.0045
2022	1.8248	0.0043	1.8248	0.0043
2023	1.6377	0.0042	1.6377	0.0042

Notes: Coefficients are standardized. Results are based on repeated simulations of the response mechanism estimated on imputed emissions.

C.4 The tilting correction

Table A9: Summary statistics (mean and standard deviation, in metric tonnes CO₂) of firm-level Scope 1 emissions under different imputation strategies: the missing-at-random baseline (non-tilted), and the MNAR correction under the fully parametric (FP) and semiparametric (SP) response models.

	$B = 300$		$B = 500$	
	Mean	St. Dev.	Mean	St. Dev.
MAR	954,567	44,110	957,284	46,054
MNAR (FP)	1,068,600	80,314	1,075,583	86,510
MNAR (SP)	1,139,399	82,840	1,139,831	84,642

D Figures



Figure A2: **Transition-firm validation: prediction error by emissions quartile.** For the 633 firms that transition into disclosure, the plot shows the prediction error (log predicted minus log subsequently-disclosed Scope 1 emissions), measured in the year before disclosure, by quartile of disclosed emissions. MNAR-corrected imputations versus Trucost vendor estimates; positive values indicate overstatement. The MNAR correction is nearly unbiased in the top quartile, where Trucost understates emissions by about 0.7 log units.